# New Enterprise Registrations

Drivers of Business Activity

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2026-03-14

# Contents


**1 Literature review** **3**

**2 Theoretical Rationale for Variable Selection and Data Sources** **4**

**3 Exploratory Data Analysis** **8**
3.1 Sample Representativeness Analysis . . . 8
3.2 Correlation Analysis . . . 8
3.3 Preliminary Regressions . . . 9

**4 The Models** **10**
4.1 Methodology and Functional Form . . . 10
4.2 Base Model with Three Business-Ready Pillars . . . 10
4.3 Addressing Multicollinearity: From Three Scores to a Weighted Average . . . 11
4.4 Pooling Pre-2024 Observations . . . 12
4.5 Individual Effects: Testing Additional Variables . . . 12
4.6 The Best Model . . . 13
4.7 Explanatory Power of Independent Variables . . . 14

**5 Model Choice and Summary** **15**
5.1 Residuals Analysis . . . 15
5.2 Predictions Decomposition . . . 15
5.3 Variable Contribution Decomposition . . . 16
5.4 Country-to-Country Distance Decomposition . . . 16

**6 Conclusion** **17**

**7 Appendix** **18**
7.1 EU South European Countries Variable Contribution Decomposition . . . 29
7.2 Other Variable Contribution Decomposition Applications . . . 31

# 1 Literature review

The question of what attracts business to a particular location has long been a central topic in regional economics and entrepreneurship studies. Traditionally, research has focused on agglomeration economies, where firms benefit from co-locating with other businesses through mechanisms of matching, sharing, and learning. However, recent scholarship suggests that a more complex interplay of factors, including population density, household characteristics, and the regulatory environment, also plays a crucial role in determining where new businesses emerge. This review synthesizes findings from key studies to explore the diverse determinants of business location and new business formation. Kopczewska, Kubara, and Kopyt (2024) challenge the traditional focus on business-to-business attraction by introducing the concept of "economies of density". The authors argue that population density itself acts as a significant attractor for firms, operating through a hierarchical causal mechanism. The mediation model confirms that the effect of population density on the location of 1st line firms is largely mediated by the presence of 2nd line firms, with approximately 96-98% of the effect being indirect. This research significantly extends business location theory by establishing population density and the resulting "convenience" of a place as a fundamental, complementary force alongside traditional agglomeration economies.

Shifting the focus from the macro-geographic region to the individual entrepreneur, Krasniqi (2009) investigates the determinants of entrepreneurship at the personal and household level. Based on a survey in Kosovo, the study finds that an individual's decision to start a business is not solely driven by regional market opportunities but is profoundly influenced by personal and household characteristics. Key determinants include prior work experience, which provides necessary skills and knowledge, and access to financial capital, often sourced from personal savings or family. This perspective complements macro-level studies by emphasizing that the "supply" of entrepreneurs is contingent on individual circumstances and resources, which act as a filter through which regional opportunities are pursued.

Focusing on the regional level, Alexandrova (2015) examines the effects of regional determinants on new business formation in Bulgaria. The study analyzes data from Bulgaria's 28 districts (NUTS-3 level) and finds that market size, measured by population and population density, is a consistently strong and positive predictor of new firm creation across all sectors. A larger population provides a greater customer base and a deeper labour pool, supporting the arguments made by Kopczewska et al. (2024) regarding the foundational role of population. Furthermore, the study identifies that existing economic structures matter; regions with a higher share of employment in services see more new firm formation in that sector, while a large agricultural sector can sometimes act as a barrier to entrepreneurship in other sectors by tying up labour and capital.

Finally, the enabling environment for business is not only economic and demographic but also regulatory. Madlener and Specht (2018) provide a crucial overview of how the regulatory framework shapes business opportunities. While their paper is broader in scope, it underscores that a stable, transparent, and predictable regulatory environment is a prerequisite for business formation and investment. Uncertainty regarding taxes, government spending, and fiscal policies can deter entrepreneurs and investors. A supportive framework – including clear property rights, efficient business registration, and consistent enforcement of contracts – lowers transaction costs and reduces the risks associated with starting a venture. This macro-institutional context interacts with the micro and regional factors discussed above.

In conclusion, the papers reviewed demonstrate that business location and new business formation are multi-faceted phenomena determined by a complex of factors operating at different levels. The research should continue to build integrated models that account for these hierarchical and complementary relationships to provide a complete picture of the dynamics that shape enterpreneurship.

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## 2 Theoretical Rationale for Variable Selection and Data Sources

Our empirical investigation is founded on the premise that the intensity of new business formation (new business density) is determined by a complex set of factors. These can be categorized into several groups: macroeconomic conditions, the quality of the institutional and regulatory environment, and the socio-demographic characteristics of the population. In this section, we justify the selection of each group of variables and describe their data sources, with a focus on 2024 as the primary period of analysis.

*Dependent Variable: New Business Density*

The dependent variable representing the level of entrepreneurial activity is the new business density rate (new_bdr). This indicator is calculated as the number of newly registered limited liability companies (LLCs) per 1,000 working-age individuals (ages 15-64). Using this relative measure allows for a correct comparison between countries with different population sizes.

The source is the World Bank Entrepreneurship Database (2024). For seven countries, due to the unavailability of current data, values from previous years (2018–2022) were used, a point explicitly noted in the analysis. In this study, we use the natural logarithm of the indicator multiplied by 100 (ln(new_bdr*100)). This transformation stabilizes the variance and allows us to interpret the coefficients of continuous predictors in terms of elasticity.

*Key Independent Variable: Quality of the Business Environment*

The central hypothesis of this paper is that a more favorable institutional environment stimulates new business creation. To test this, we utilize data from the new World Bank report, "Business Ready (B-READY) 2024" , which succeeded the Doing Business ranking. We employ the three pillars of this index:

Regulatory Framework (s_reg): Assesses the quality of laws and regulations governing business entry and operation (e.g., transparency of requirements, minority investor protections).

Public Services (s_pub): Reflects the availability and quality of the digital and physical infrastructure essential for businesses (e.g., online business registration services, access to electricity, quality of land cadasters).

Operational Efficiency (s_opr): Measures the efficiency and ease with which procedures outlined in the regulatory framework are implemented in practice (e.g., speed of registration, permit processing, contract enforcement).

We hypothesize that higher scores in each of these pillars lead to increased entrepreneurial activity. To mitigate multicollinearity and simplify the model, we constructed a weighted average score (s_wavg) based on the coefficients obtained from a regression model using the three separate pillar scores.

*Macroeconomic Control Variables*

Economic theory suggests that the level of entrepreneurial activity is strongly dependent on market size and the population's purchasing power.

GDP per adult (gdp_padult): We use this indicator as a proxy for the level of economic development and prosperity. A higher GDP per adult signifies a larger market and greater potential profit for new businesses. The expected sign of the coefficient is positive. Data source: World Bank Open Data (2024).

Adult population (adults): This variable serves as a control because it is the denominator in the calculation of our dependent variable. Including it in logarithmic form, according to the rules of logarithms, allows us to eliminate the "spurious correlation" problem caused by the common denominator and to estimate the effect of economic scale. The expected sign of the coefficient is negative. This is because, holding total GDP (and thus GDP per adult) constant, an increase in the adult population means more potential entrepreneurs competing for the same "economic pie," which intensifies competition and can complicate market entry, while putting additional pressure on the public services.

Consumer Price Index (CPI): Inflation reflects the overall macroeconomic stability of an economy and may influence entrepreneurial activity through several channels. High and volatile inflation increases uncertainty regarding future costs and revenues, which may discourage investment and the creation of new firms. At the same time, moderate inflation can sometimes stimulate economic activity by increasing nominal revenues. In general, however, macroeconomic instability associated with high inflation is expected to negatively affect entrepreneurial activity. Source: World Bank Open Data (2024).

*Barriers to Entry*

To test the impact of administrative barriers, we use two classic indicators from the Doing Business archive: Paid-in Minimum Capital (cap_req): Expressed as a percentage of income per capita. High minimum capital requirements can deter potential entrepreneurs who lack access to financing (Krasniqi, 2009). The expected sign is negative.

Cost of Starting a Business (cost): Expressed as a percentage of income per capita. The more expensive the registration procedure, the lower the incentive for formal business creation. The expected sign is negative. Data for these variables are sourced from the Doing Business archive (2020 data), as this represents the most complete and relevant dataset for these slowly changing indicators.

Remote business registration (remote): The ability to register a business remotely through digital or online procedures may significantly reduce administrative barriers to entrepreneurship. Digital registration systems lower transaction costs, reduce the time required to establish a firm, and simplify bureaucratic procedures for entrepreneurs. Countries that allow remote business registration may therefore experience higher rates of formal business creation. The expected relationship with entrepreneurial activity is positive. Source: World Bank Business Ready (B-READY) indicators.

*Socio-Demographic and Structural Factors*

The "economies of density" theory (Kopczewska et al., 2024) and research on regional determinants (Alexandrova, 2015) highlight the

importance of population structure and concentration.
Urbanisation (urbanisation): The share of the population living in urban areas. Cities have a higher density of economic agents, facilitating knowledge spillovers and the matching of firms with workers and customers. The expected sign is positive. Source: World Bank Open Data (2024).
Working-Age Population (pct_adult): The proportion of the population aged 15-64. A larger share of working-age individuals increases the pool of potential entrepreneurs and employees. The expected sign is positive. Source: World Bank Open Data (2024).
Unemployment rate (unemployment): The unemployment rate may influence entrepreneurial activity through two opposing mechanisms. On the one hand, high unemployment can stimulate necessity-driven entrepreneurship, as individuals who are unable to find paid employment may turn to self-employment as an alternative source of income. On the other hand, high unemployment may also reflect weak economic conditions and limited demand, which could discourage new firm creation. Therefore, the expected sign of the coefficient is theoretically ambiguous. Source: World Bank Open Data (2024).
Education (education): Human capital is widely recognized as an important determinant of entrepreneurial activity. A higher level of education increases individuals' managerial skills, financial literacy, and ability to identify and exploit market opportunities. More educated populations may therefore be more likely to engage in opportunity-driven entrepreneurship and to establish formally registered firms. The expected sign of the coefficient is positive. Source: United Nations Development Programme (UNDP), Human Development Report (2025).
Income inequality (gini): Income inequality may influence entrepreneurial activity through multiple channels. On the one hand, higher inequality may increase incentives for entrepreneurship by creating strong financial rewards for successful business ventures. On the other hand, high inequality may reflect limited access to capital and opportunities for large segments of the population, potentially restricting the pool of potential entrepreneurs. Due to these competing mechanisms, the expected sign of the coefficient is theoretically ambiguous. Source: World Bank Open Data (2024). Home ownership rate (home_own): The home ownership rate may influence entrepreneurship through household wealth and financial stability. Individuals who own property often have greater access to collateral, which may facilitate borrowing and financing for new business ventures. Home ownership may therefore lower financial constraints faced by potential entrepreneurs and increase the likelihood of starting a new firm. A positive relationship between home ownership and entrepreneurial activity is therefore expected. Source: OECD Housing Database (2024).
Population density (pop_dens): Population density captures the spatial concentration of economic activity and may influence entrepreneurial dynamics through agglomeration effects. Densely populated areas typically provide larger markets, better infrastructure, and stronger knowledge spillovers, which may facilitate the creation and growth of new businesses. Higher population density is therefore expected to be positively associated with new firm formation. Source: World Bank Open Data (2024).
Population growth (pop_growth): Population growth reflects the expansion of potential markets and labor supply in an economy. Growing populations may increase demand for goods and services, creating new market opportunities for entrepreneurs. In addition, population growth may expand the pool of potential business founders. Therefore, higher population growth is generally expected to be positively related to entrepreneurial activity. Source: World Bank Open Data (2024).
Female population share (pct_female): The gender composition of the population may influence entrepreneurial activity through labor market participation and gender-specific barriers to entrepreneurship. A larger female population may increase the potential pool of entrepreneurs, particularly in countries where female labor force participation and access to financial resources are improving. At the same time, structural barriers and social norms may still limit female entrepreneurship in some economies. Therefore, the relationship between the share of the female population and new business formation may vary across institutional contexts. Source: World Bank Open Data (2024).
Share of minors in the population (pct_minor): The age structure of the population may affect entrepreneurial dynamics through its impact on labor supply and economic dependency ratios. A higher share of minors typically reflects a younger population structure, which may influence long-term labor market dynamics and future entrepreneurial potential. However, in the short run, a high dependency ratio may reduce available resources for investment and business creation. The expected relationship with entrepreneurial activity is therefore theoretically ambiguous. Source: World Bank Open Data (2024).

*Institutional and Country Characteristics*

Income Group (g_income): A categorical variable (Low, Lower-middle, Upper-middle, High income) classifying countries by their Gross National Income (GNI) per capita. This variable acts as a proxy for a wide range of unobserved country-specific characteristics related to the overall level of development. The coefficients will show the average difference in entrepreneurial activity between income groups, holding other factors constant. Source: World Bank Country Classification (2024).
Corruption perception (corruption): The level of perceived corruption reflects the broader institutional environment in which businesses operate. High levels of corruption increase transaction costs, create uncertainty, and may discourage entrepreneurs from entering the formal sector. Conversely, lower corruption levels improve trust in public institutions and reduce informal barriers to market entry. Therefore, we expect a positive relationship between lower corruption and higher rates of new business formation. Data source: Transparency International, Corruption Perceptions Index (2024).
Democracy index (democracy): The level of democracy reflects the broader political and institutional environment in which economic activity takes place. Democratic institutions are typically associated with stronger protection of property rights, greater government accountability, and more transparent policymaking. These features reduce uncertainty for entrepreneurs and investors and may encourage the creation of new firms. Therefore, a positive relationship between the level of democracy and entrepreneurial activity can be expected. Source: Economist Intelligence Unit Democracy Index (2024).

Offshore financial center classification (is_ofc): Some jurisdictions function as offshore financial centers, characterized by favorable regulatory frameworks, low taxation, and a strong orientation toward international financial services. These institutional features may attract a large number of registered firms, including holding companies and special-purpose entities. As a result, offshore financial centers may display unusually high levels of firm registrations compared to the size of their domestic economies. This variable is included to control for structural differences between offshore jurisdictions and standard national economies. Source: IMF Offshore Financial Center classification.

*Taxation and Fiscal Environment*

Corporate tax rate (corp_tax): Corporate taxation influences the expected profitability of entrepreneurial activity. Higher corporate tax rates reduce the net returns from business operations and may discourage individuals from starting new firms or encourage them to remain in the informal sector. Conversely, lower corporate tax rates increase expected after-tax profits and may stimulate formal business creation. The expected sign of the coefficient is therefore negative. Source: Tax Foundation (2024).

International Tax Competitiveness Index (tax_comp): In addition to statutory tax rates, the overall structure and competitiveness of a country's tax system may affect entrepreneurial activity. The International Tax Competitiveness Index evaluates how neutral, transparent, and investment-friendly a country's tax system is compared to others. More competitive tax systems reduce distortions and administrative burdens, making it easier for businesses to operate and expand. Therefore, higher values of the index are expected to be associated with greater entrepreneurial activity. Source: Tax Foundation (2024).

Overall, the selected variables reflect the main theoretical determinants of entrepreneurial activity discussed in the literature, including macroeconomic conditions, institutional quality, regulatory barriers, and demographic characteristics. Together, they form the empirical framework used to analyze cross-country differences in new business formation.

The complete list of variable designations can be found in Table 1.

Table 1. Complete List of Variables

| Variable | Description |
|---|---|
| code | Country code |
| country | Country name |
| w_region | World region according to World Bank |
| block | Economic block |
| g_income | Income group according to World Bank |
| g_lending | Lending category according to World Bank |
| adults | Adult population |
| year | Year |
| new_llcs | Number of new Limited Liability Companies registered this year |
| new_bdr | New business density rate: number of new LLCs per 1000 working-age population |
| ln100x_new_bdr | Ln of (New business density rate x 100): the lowest BDI in the dataset is a 2nd order decimal and needs multiplication by 100 for the ln to work properly |
| is_ofc | Included in the Eurostat List of offshore financial centres |
| pre2024 | New LLC registrations data collected before 2024 |
| remote | Business can be registered remotely |
| s_avg | Average Business-Ready score |
| s_wavg | Weighted Average Business-Ready score: the weights are coefficients acquired from the model with the 3 separate scores |
| s_reg | Pillar 1 Regulatory Framework |
| s_pub | Pillar 2 Public Services |
| s_opr | Pillar 3 Operational Efficiency |
| cap_req | Minimum capital requirements to start a business, as a % of average income |
| cost | Cost of starting a business, as a % of average income |
| start_time | Time required to start a business, in calendar days |
| avg_income | Average income in US$ |
| gdp | GDP (current US$) |
| gdp_pcap | GDP per capita (current US$) |
| gdp_padult | GDP per adult (current US$) |
| ln100kx_gdp_padult | Ln of (GDP per adult (current US$) x 100,000): GDP per adult coerced to a denominator identical to the one in Ln of (New business density rate x 100) |
| corp_tax | Corporate Tax Rate |
| tax_comp | International tax competitiveness index |
| gini | Gini coefficient |
| democracy | Democracy Index |
| corruption | Corruption Perception Index |
| education | Average years of schooling among adults |
| inflation | Inflation, consumer prices (annual %) |
| unemployment | Unemployment, total (% of total labor force) (national estimate) |
| home_own | Home ownership rate |
| pop_growth | Population growth (annual %) |
| pop_dens | Population density (people per sq. km of land area) |
| urbanisation | Urban population (% of total population) |
| pct_adut | Population ages 15-64 (% of total population) |
| pct_minor | Population ages 0-14 (% of total population) |
| pct_female | Population, female (% of total population) |

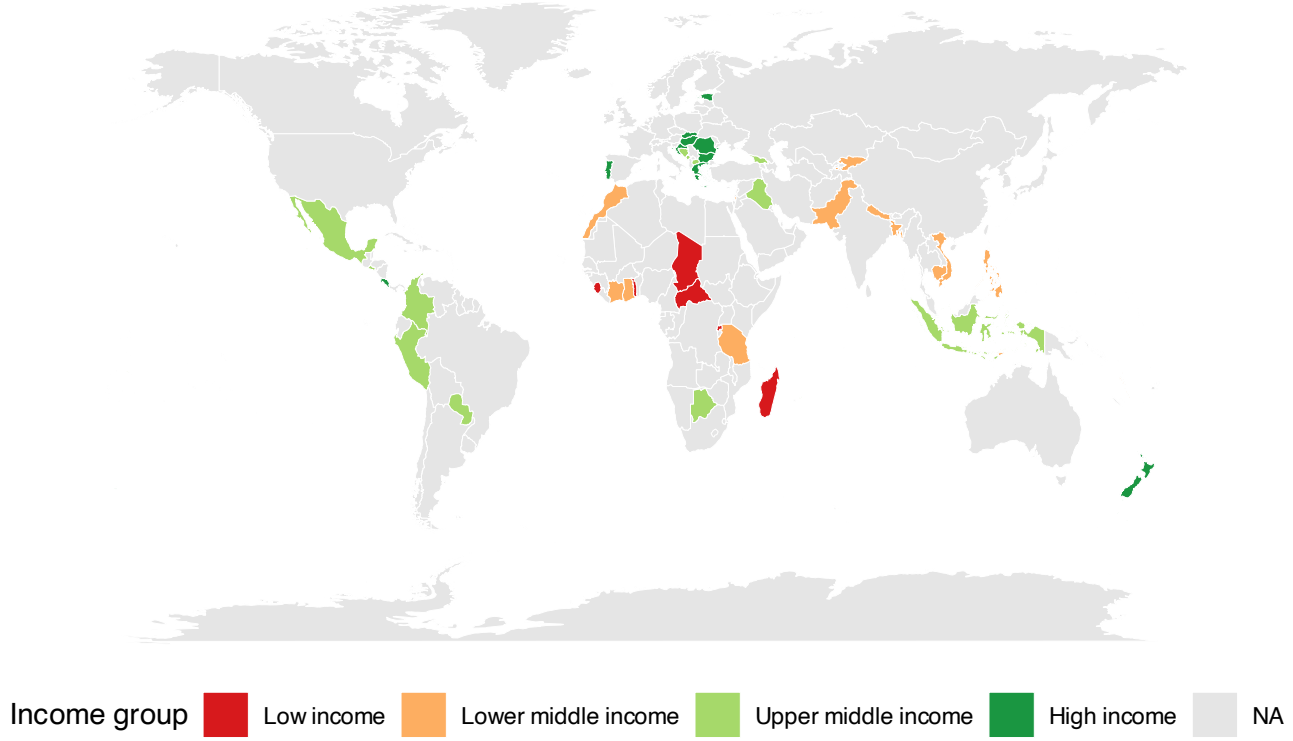
# 3 Exploratory Data Analysis

## 3.1 Sample Representativeness Analysis

The final sample for empirical analysis consists of 50 countries. A key constraint in sample formation was the availability of data for the Business Ready index for the year 2024. Consequently, the world's largest economies (such as the USA, China, Germany, the UK, France, Russia, and others) are not included in the sample. This limitation must be considered when interpreting the results and extrapolating the findings to these countries. Nevertheless, in terms of geographic representation and income level distribution, the sample is reasonably balanced.

As shown in Figure 1, the sample extensively features countries from Europe and Central Asia (e.g., Bulgaria, Greece, Croatia), Sub-Saharan Africa (Botswana, Tanzania, Lesotho), Latin America and the Caribbean (Barbados, Mexico), as well as East Asia and the Pacific (Cambodia, Vietnam, New Zealand). This diversity allows for cross-country comparisons and enables the identification of patterns that may hold true across different world regions (see Figures 2 and 3 in Appendix).

Figure 1. Geographic Distribution of Countries in the Sample




An analysis of the countries' distribution by World Bank income groups reveals that the sample includes states from all four categories. It comprises both low-income countries (e.g., Afghanistan, Tanzania) and high-income countries (New Zealand, Portugal, Croatia). This provides the necessary variation to assess the impact of the economic development level on entrepreneurial activity and to justify the inclusion of the corresponding control variable (g_income).

## 3.2 Correlation Analysis

To understand the underlying relationships between variables, we examine the correlation matrix (see Figure 4 in Appendix). The heatmap reveals several important patterns. Variables showing strong positive correlations with new business density include GDP per capita (and GDP per adult), the Business-Ready scores across all three pillars (regulatory framework, public services, operational efficiency. This provides initial support for our theoretical expectations that economic prosperity, institutional quality, and population concentration foster entrepreneurial activity.

Conversely, variables with strong negative correlations with the dependent variable are population density, cost of starting a business, time required to start a business. These findings align with the barriers-to-entry literature (Krasniqi, 2009), suggesting that administrative burdens indeed discourage new business formation. Several variables exhibit low correlation with new business density: inflation, unemployment, and the percentage of female population. Based on this preliminary evidence, these variables are excluded from subsequent model specifications to preserve degrees of freedom given the limited sample size.

### 3.3 Preliminary Regressions

Before proceeding to multivariate analysis, we examine the bivariate relationships between our key variables of interest and the dependent variable. The first plot (see Figure 5 and 6) shows the relationship between logged GDP per adult and logged new business density. A clear positive association emerges, consistent with the hypothesis that larger markets and greater purchasing power stimulate entrepreneurial activity. The second plot displays the relationship between the average Business-Ready score and logged new business density, also revealing a positive association that supports the institutional quality hypothesis.

Figure 5. Preliminary Regression of New Business Density Rate on GDP per Adult

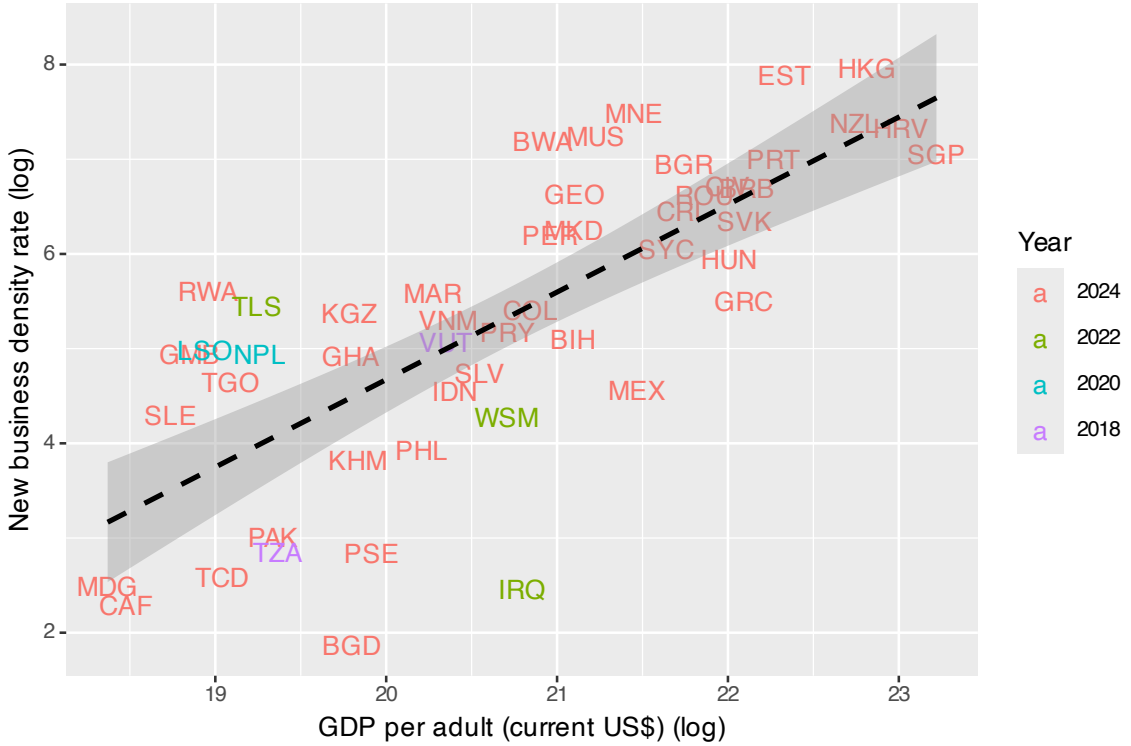
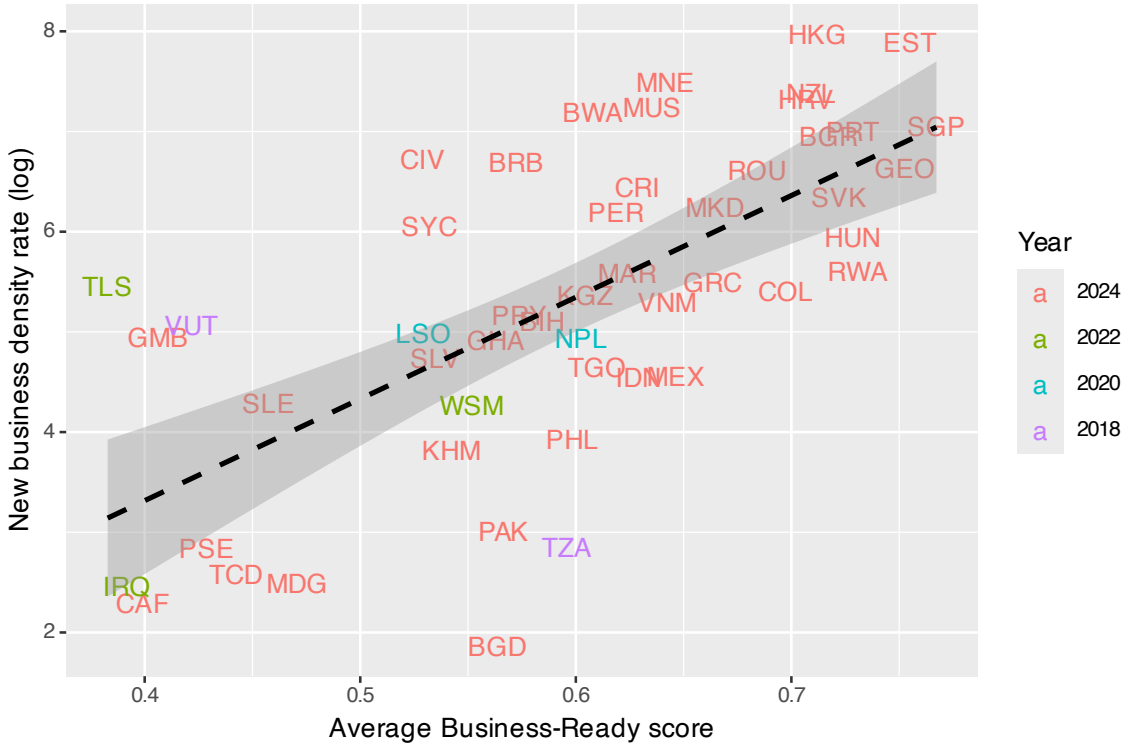



Figure 6. Preliminary Regression of New Business Density Rate on Average Business-Ready Score



# 4 The Models

## 4.1 Methodology and Functional Form

Following the theoretical framework established in Section 2, we specify a series of regression models to explain cross-country variation in new business density. The dependent variable is specified as the natural logarithm of new business density multiplied by 100:

$$\ln\left(New\ BDR_i \times 100\right)$$

This transformation serves multiple purposes. First, multiplying by 100 ensures that even the smallest values in the dataset (which are second-order decimals) exceed unity, making the logarithmic transformation mathematically feasible. Second, the logarithmic transformation stabilizes the variance of the dependent variable, addressing potential heteroskedasticity. Third, and most importantly, this specification allows us to exploit the properties of logarithms to eliminate spurious correlation arising from the common denominator problem.

Recall that new business density is defined as new LLC registrations per 1,000 working-age individuals:

$$New\ BDR_i = \frac{New\ LLCs_i \times 1000}{Adult\ population_i}$$

Taking logs and multiplying by 100 allows the following specification:

$$\ln\left(\frac{New\ LLCs_i \times 1000}{Adult\ population_i} \times 100\right) =$$
$$\beta_0 + \beta_1 \ln\left(\frac{GDP_i}{Adult\ population_i} \times 1000 \times 100\right) + \gamma \ln\left(Adult\ population_i\right) + \beta_2 X_{i,2} + \cdots + \beta_k X_{i,k} + \varepsilon_i$$

Expanding the left-hand side and the GDP per adult term using quotient rule ($\log\left(\frac{a}{b}\right) = \log a - \log b$):

$$\ln\left(\left(Number\ of\ new\ LLCs_i \times 1000\right) \times 100\right) - \ln\left(Adult\ population_i\right) =$$
$$\beta_0 + \beta_1 \left[\ln\left(GDP_i \times 1000 \times 100\right) - \ln\left(Adult\ population_i\right)\right] + \gamma \ln\left(Adult\ population_i\right) + \beta_2 X_{i,2} + \cdots + \beta_k X_{i,k} + \varepsilon_i$$

This formulation reveals that including adult population on both sides allows us to estimate the pure scale effect ($\gamma$) holding total GDP (and thus GDP per adult) constant. A negative coefficient on adult population would indicate that, for a given economic pie, more potential entrepreneurs intensify competition, complicate market entry and put strain on public services.

For interpretability, all variables expressed as percentages are converted to fractions (e.g., 83.9% urbanisation becomes 0.839). This allows coefficients for such variables to be interpreted as the percentage change in the dependent variable for a one percentage point increase in the predictor.

## 4.2 Base Model with Three Business-Ready Pillars

We begin by estimating a model that includes the three separate Business-Ready pillar scores alongside the macroeconomic controls:

$$\ln\left(New\ BDR_i \times 100\right) = \beta_0 + \beta_1 \ln\left(GDP\ per\ adult_i \times 1000 \times 100\right) + \gamma \ln\left(Adult\ population_i\right) +$$
$$+\beta_2\ Regulatory\ Framework_i + \beta_3\ Public\ Services_i + \beta_4\ Operational\ Efficiency_i + \varepsilon_i$$

The results (see Table 2) show that all three pillars have positive coefficients, though only public services achieves conventional significance levels. GDP per adult and adult population perform as expected, with the latter confirming the competition-intensifying effect. However, diagnostic tests reveal a potential issue: variance inflation factors indicate problematic multicollinearity, particularly for the public services score (VIF = 5.72). This is unsurprising given that the three pillars are designed to capture related aspects of the business environment and are naturally correlated. However, multicollinearity is reduced to insignificance when the scores are averaged, be it weighted or arithmetic average. Therefore, we proceed to a model with a weighted average of the 3 scores (the weights are coefficients acquired from the model with the 3 separate scores).

Table 2. From Three Scores to a Weighted Average

| | Mult. | Arithm. Avg. | Weighted Avg. |
|---|---|---|---|
| (Intercept) | 2.048 | -1.006 | 2.048 |
| | (3.447) | (2.792) | (3.005) |
| log(gdp_padult * 100000) | 0.277+++ | 0.359** | 0.277* |
| | (0.141) | (0.132) | (0.136) |
| log(adults) | -0.428*** | -0.381*** | -0.428*** |
| | (0.078) | (0.072) | (0.073) |
| s_reg | 2.172 | | |
| | (2.474) | | |
| s_pub | 4.852** | | |
| | (1.793) | | |
| s_opr | 0.472 | | |
| | (1.517) | | |
| s_avg | | 7.992*** | |
| | | (1.579) | |
| s_wavg | | | 7.496*** |
| | | | (1.376) |
| Num.Obs. | 50 | 50 | 50 |
| R2 | 0.774 | 0.761 | 0.774 |
| R2 Adj. | 0.748 | 0.745 | 0.759 |
| AIC | 199.2 | 197.9 | 195.2 |
| BIC | 212.6 | 207.5 | 204.8 |
| Log.Lik. | -56.922 | -58.290 | -56.922 |
| F | 30.094 | 48.827 | 52.436 |
| RMSE | 0.76 | 0.78 | 0.76 |

+ p < 0.3, ++ p < 0.2, +++ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

## 4.3 Addressing Multicollinearity: From Three Scores to a Weighted Average

**1. Variance Inflation Factor Analysis**

To address the multicollinearity concern, we consider two alternative specifications: an arithmetic average of the three scores and a weighted average. The arithmetic average is straightforward, and the weighted average is constructed using the coefficients estimated from the three-score model as weights:

$$Weighted\ Avg.\ Score = 2.172\ Regulatory\ Framework_i + 4.852\ Public\ Services_i + 0.472\ Operational\ Efficiency_i$$

VIF diagnostics confirm that both averaging approaches successfully mitigate multicollinearity. In the arithmetic average model, all VIF values fall below the conventional threshold of 5 (ranging from 1.27 to 2.20). The weighted average model similarly shows acceptable VIF values (1.39 to 2.48), confirming that the averaging strategy resolves the multicollinearity problem.

**2. Model Comparison: ANOVA Tests**

To determine whether the simpler average-score models sacrifice explanatory power, we conduct ANOVA F-tests comparing the full three-score model against each restricted model. Comparing the three-score model against the arithmetic average model yields an F-statistic with p = 0.3, indicating no significant loss of fit. Comparing against the weighted average model yields p = 1.0, strongly confirming that the restricted model performs equivalently to the more complex specification.

**3. Information Criteria Comparison**

The Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) provide a more nuanced comparison that balances goodness of fit against model parsimony. The three-score model yields AIC = 199.2 and BIC = 212.6. The arithmetic average model improves slightly (AIC = 197.9, BIC = 207.5). The weighted average model performs best, with the lowest values for both criteria (AIC = 195.2, BIC = 204.8). The adjusted $R^2$ also favors the weighted average model (0.759) compared to the arithmetic average (0.745) and three-score model (0.748). Based on these diagnostics, we select the weighted average score as our preferred measure of institutional quality and proceed with it in subsequent analysis.

## 4.4 Pooling Pre-2024 Observations

**1. The Issue**

Seven countries in our sample have new business density data from years prior to 2024 (ranging from 2018 to 2022). These are predominantly lower-middle income economies with low lending ratings: Iraq (2022), Lesotho (2020), Nepal (2020), Samoa (2022), Tanzania (2018), Timor-Leste (2022), and Vanuatu (2018). Including these observations introduces a measurement error, as the dependent variable is not contemporaneous with the independent variables (all measured in 2024). However, excluding them would further reduce an already limited sample.

**2. Chow Test for Structural Break**

To empirically assess whether these pre-2024 observations can be pooled with the 2024 data, we conduct a Chow test. This test compares a restricted model (no interaction with the pre2024 dummy) against an unrestricted model that allows all coefficients to differ between the two groups. The analysis of variance yields an F-statistic of 3.81 with $p = 0.0099$, indicating that the relationship between the independent variables and new business density is structurally different for the pre-2024 observations at conventional significance levels.

**3. Linear Restriction Test for Coefficient Stability**

However, the Chow test's significance could be driven by differences in intercepts or a subset of coefficients rather than instability in the core relationships. We therefore conduct a more focused linear restriction (Wald) test, examining whether the coefficients on our key variables – GDP per adult, adult population, and the weighted average score – differ significantly from the benchmark model. The test fails to reject the null hypothesis ($p = 0.97$), suggesting that while the overall model structure may differ, the core coefficients are remarkably stable.

**4. Decision to Pool**

Visual inspection of the pre-2024 observations reveals that these seven economies appear different from the rest of the sample not as systematic outliers but as a particular cluster sharing similar characteristics. Their prediction errors in the final model are in line with the majority of observations, further supporting their inclusion. Given the results of the linear restriction test, the visual evidence, and the need to preserve sample size, we proceed with pooling these observations while acknowledging this limitation in interpreting results.

## 4.5 Individual Effects: Testing Additional Variables

With the core model established – log GDP per adult, log adult population, and weighted average Business-Ready score – we systematically test the incremental contribution of each potential control variable identified in Section 2. Given the limited sample size ($n = 50$), we adopt a liberal inclusion criterion, considering variables with $p < 0.2$ or even $p < 0.3$ in individual tests to avoid overlooking potentially important relationships. Table 3 (see Appendix) summarizes the results of these individual effects models.

Several variables emerge as potentially important based on these individual tests:
**Cost of starting a business** (cost) shows a consistently negative coefficient significant at the 5% level, confirming that higher registration costs discourage formal entrepreneurship. Each percentage point increase in cost (as a share of income per capita) is associated with approximately a 0.01% decrease in new business density.
**Minimum capital requirements** (cap_req) also exhibits the expected negative sign and achieves marginal significance ($p < 0.1$) in individual tests, though significance attenuates when other variables are included.
**Urbanisation** (urbanisation) demonstrates a strong positive relationship ($p < 0.1$), with each percentage point increase in urban population share associated with approximately a 1.27% increase in new business density. This supports the "economies of density" hypothesis (Kopczewska et al., 2024).
**Working-age population share** (pct_adult) shows a substantial positive effect ($p < 0.1$), with each percentage point increase associated with roughly a 5.3% increase in new business density. This confirms that the demographic structure matters for entrepreneurial potential.
**Corruption perceptions** (corruption) and **democracy index** both show positive associations with marginal significance ($p < 0.2$), suggesting that institutional quality beyond the business-specific regulatory framework may matter.
**Income group** (g_income) dummies show a curious pattern: contrary to simple intuition, higher income categories are associated with lower new business density after controlling for GDP per adult and other factors. This may reflect that in high-income countries, entrepreneurial opportunities are relatively saturated compared to emerging economies, or that opportunity entrepreneurship differs fundamentally from necessity entrepreneurship.

Variables showing no consistent relationship include inflation, unemployment, home ownership rate, population growth, population density, gender composition, and the offshore financial center indicator. These are excluded from subsequent models to preserve parsimony.

## 4.6 The Best Model

**1. Model Specification**

Based on the individual effects analysis and theoretical considerations, we specify the following best model:

$$\ln(New\ BDR_i \times 100) = \beta_0 + \beta_1 \ln(GDP\ per\ adult_i \times 1000 \times 100) + \gamma \ln(Adult\ population_i) +$$
$$+\beta_2\ Weighted\ Avg.\ Score_i + \beta_3\ Min.capital\ requirements + \beta_4\ Cost\ of\ starting +$$
$$+\beta_5\ Urbanisation + \beta_6\ Working - age\ population\ share + \beta_7\ Lower - middle\ income\ dummy +$$
$$+\beta_8\ Upper - middle\ income\ dummy\ + \beta_9\ High\ income\ dummy + \varepsilon_i$$

**2. Coefficients Interpretation**

Table 4 presents the estimation results for the best model.

Table 4. Best Model Summary

| | Best |
|---|---|
| (Intercept) | -2.780 |
| | (4.692) |
| log(gdp_padult * 100000) | 0.467* |
| | (0.223) |
| log(adults) | -0.489*** |
| | (0.077) |
| s_wavg | 6.543*** |
| | (1.455) |
| cap_req | -0.449 |
| | (0.954) |
| cost | -0.008++ |
| | (0.005) |
| urbanisation | 1.299+++ |
| | (0.718) |
| pct_adut | 4.597++ |
| | (3.060) |
| g_incomeLower middle income | -0.655++ |
| | (0.450) |
| g_incomeUpper middle income | -1.345* |
| | (0.603) |
| g_incomeHigh income | -1.884* |
| | (0.821) |
| Num.Obs. | 50 |
| R2 | 0.843 |
| R2 Adj. | 0.803 |
| AIC | 190.9 |
| BIC | 213.8 |
| Log.Lik. | -47.764 |
| F | 20.963 |
| RMSE | 0.63 |
| + p < 0.3, ++ p < 0.2, +++ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001 | |

The coefficients can be interpreted as follows:

**GDP per adult**: A 1% increase in GDP per adult results in a 0.467% increase in new business density ($p < 0.05$). This elasticity is somewhat higher than the average of 0.266% across the 21 individual effects models, suggesting that controlling for other factors strengthens the estimated market size effect. This supports the hypothesis that greater purchasing power and larger markets stimulate entrepreneurial activity.

**Adult population**: A 1% simultaneous increase in adult population and total GDP (so that GDP per adult remains constant) results in a 0.489% decrease in new business density ($p < 0.001$). This is slightly more negative than the average of -0.437% across models, confirming that for a given economic pie, more potential entrepreneurs intensify competition, complicate market entry and put strain on public services. This coefficient captures the pure scale effect hypothesized in Section 2.

**Weighted average Business-Ready score**: A one percentage point increase (0.01) in the weighted average score leads to a 6.543% increase in new business density ($p < 0.001$). While slightly lower than the average effect of 7.295% across models, this remains a substantively large and highly significant effect. It confirms our central hypothesis: a more favorable institutional environment – encompassing regulatory framework, public services, and operational efficiency – substantially stimulates new business creation.
**Minimum capital requirements**: A one percentage point increase in minimum capital requirements (as a share of income per capita) is associated with a 0.449% decrease in new business density. Though not statistically significant in the full model ($p > 0.3$), the coefficient retains the expected negative sign and magnitude. Given its theoretical importance and significance in individual tests, we retain it for theoretical completeness.
**Cost of starting a business**: A one percentage point increase in registration costs reduces new business density by 0.008% ($p < 0.2$). This effect, while small in magnitude, is consistent with expectations and aligns with the barriers-to-entry literature (Krasniqi, 2009).
**Urbanization**: A one percentage point increase in the share of population living in urban areas increases new business density by 1.299% ($p < 0.1$). This supports Kopczewska et al.'s (2024) "economies of density" theory: cities facilitate knowledge spillovers and matching between firms, workers, and customers.
**Working-age population share**: A one percentage point increase in the proportion of the population aged 15-64 increases new business density by 4.597% ($p < 0.2$). This confirms that a larger pool of potential entrepreneurs and employees enhances entrepreneurial activity.
**Income group**: Relative to low-income countries, lower-middle income countries show 0.655% lower new business density, upper-middle income countries 1.345% lower, and high-income countries 1.884% lower (all $p < 0.2$ for the latter two). This pattern, which persists despite controlling for GDP per adult and institutional quality, suggests that entrepreneurship may play a different role across development stages – perhaps reflecting the transition from necessity-driven to opportunity-driven entrepreneurship, or saturation effects in mature economies. Notably, including income group dummies improves the significance of other variables, suggesting that it absorbs important cross-country heterogeneity.

#### 3. Multicollinearity Assessment

Given the inclusion of conceptually related variables – particularly GDP per adult and income group dummies – we examine variance inflation factors for the best model. Several variables show elevated VIF values: GDP per adult (8.14), high-income group dummy (13.40), and upper-middle income group dummy (7.23). These values exceed conventional thresholds, raising concerns about coefficient precision.

To assess whether multicollinearity critically affects our estimates, we conduct linear restriction tests comparing each coefficient in the best model to the average of that coefficient across all individual effects models (see Table 5 in Appendix). The null hypothesis is that the coefficient equals this benchmark value. All tests yield high p-values (ranging from 0.228 to 0.973), indicating that despite elevated VIFs, our coefficients are not statistically different from the more stable benchmark estimates. This mitigates concerns about critical estimation errors, though we acknowledge that standard errors may be inflated and significance levels should be interpreted cautiously.

#### 4. Functional Form Specification

The Ramsey RESET test examines whether the model suffers from functional form misspecification by testing whether powers of the fitted values have explanatory power when added to the model. The test yields $F = 1$ with $p = 0.3$, failing to reject the null of correct functional specification. This provides confidence that our linear-in-parameters specification with logged transformations is appropriate.

### 4.7 Explanatory Power of Independent Variables

#### 1. Semi-Partial $R^2$

To assess each variable's unique contribution to explaining variance in new business density, we compute semi-partial (part) $R^2$ coefficients. This measures how much the overall $R^2$ would decrease if a given variable were removed from the model, holding all others constant. The analysis (see Figure 7 in Appendix) reveals that all predictors except minimum capital requirements make profound and unique contributions. The weighted average Business-Ready score shows the largest unique explanatory power, followed by adult population (the competition effect) and the income group dummies. This reinforces the centrality of institutional quality in understanding cross-country variation in entrepreneurial activity.

#### 2. Standardized Coefficients

To compare the relative strength of predictors measured on different scales, we estimate a standardized version of the best model (see Figure 8 in Appendix). The high-income group dummy, GDP per adult, and the weighted average score emerge as the most influential predictors in terms of standardized magnitude, each with coefficients exceeding 0.5 in absolute value. The confidence intervals around these estimates confirm their statistical importance while acknowledging some uncertainty due to multicollinearity.

# 5 Model Choice and Summary

Based on the preceding analysis, we select the weighted average model including urbanization, working-age population share, minimum capital requirements, cost of starting a business, and income group dummies as our preferred specification. This model achieves the best balance of explanatory power ($R^2 = 0.843$, adjusted $R^2 = 0.803$), parsimony (AIC = 190.9, BIC = 213.8), and theoretical coherence. The core hypothesis regarding institutional quality is strongly supported: the weighted average Business-Ready score remains highly significant with a substantively large coefficient across all specifications.

The model confirms that entrepreneurial activity is shaped by a complex interplay of factors: market size (GDP per adult) attracts entry, but competition among potential entrepreneurs (adult population) tempers this effect; institutional quality strongly stimulates formation; demographic structure (urbanization and working-age share) provides the human capital base; and administrative barriers (cost and capital requirements) discourage formalization. The persistent income group effects, even after controlling for these factors, suggest that broader development context matters in ways not fully captured by our measured variables.

## 5.1 Residuals Analysis

**Normality Assessment**

The ordinary least squares estimator relies on the assumption of normally distributed errors for valid inference in small samples. We assess this using the Shapiro-Wilk test, which examines whether the residuals deviate from a normal distribution. The test statistic $W \approx 1$ with $p = 0.7$ fails to reject the null hypothesis of normality. Visual inspection of the Q-Q plot (see Figure 9 in Appendix) confirms that the residuals closely follow the diagonal line, with no concerning departures in the tails. The normality assumption is therefore satisfied.

**Heteroskedasticity Assessment**

The Breusch-Pagan test examines whether the variance of residuals is constant across observations – the assumption of homoskedasticity. The studentized version of the test yields BP = 8 with df = 10 and $p = 0.6$, failing to reject the null of constant variance. A plot of fitted values against residuals (see Figure 10 Appendix) visually confirms this result: the residuals display no systematic pattern or funnel shape, instead appearing randomly scattered around zero with roughly constant spread. The homoskedasticity assumption is therefore satisfied.

## 5.2 Predictions Decomposition

**Prediction Quality and Ranking Accuracy**

The best model achieves strong in-sample predictive performance, with multiple $R^2 = 0.843$ and adjusted $R^2 = 0.803$. The mean percentage error of -0.232 indicates a slight overall tendency to underpredict, while the mean absolute percentage error of 0.596 quantifies average prediction error magnitude. A scatter plot of predicted versus actual values (see Figure 11) shows close alignment around the 45-degree line, with no evidence of systematic bias. Crucially, the seven pre-2024 observations (color-coded in the plot) are not systematic outliers, confirming the validity of pooling them.

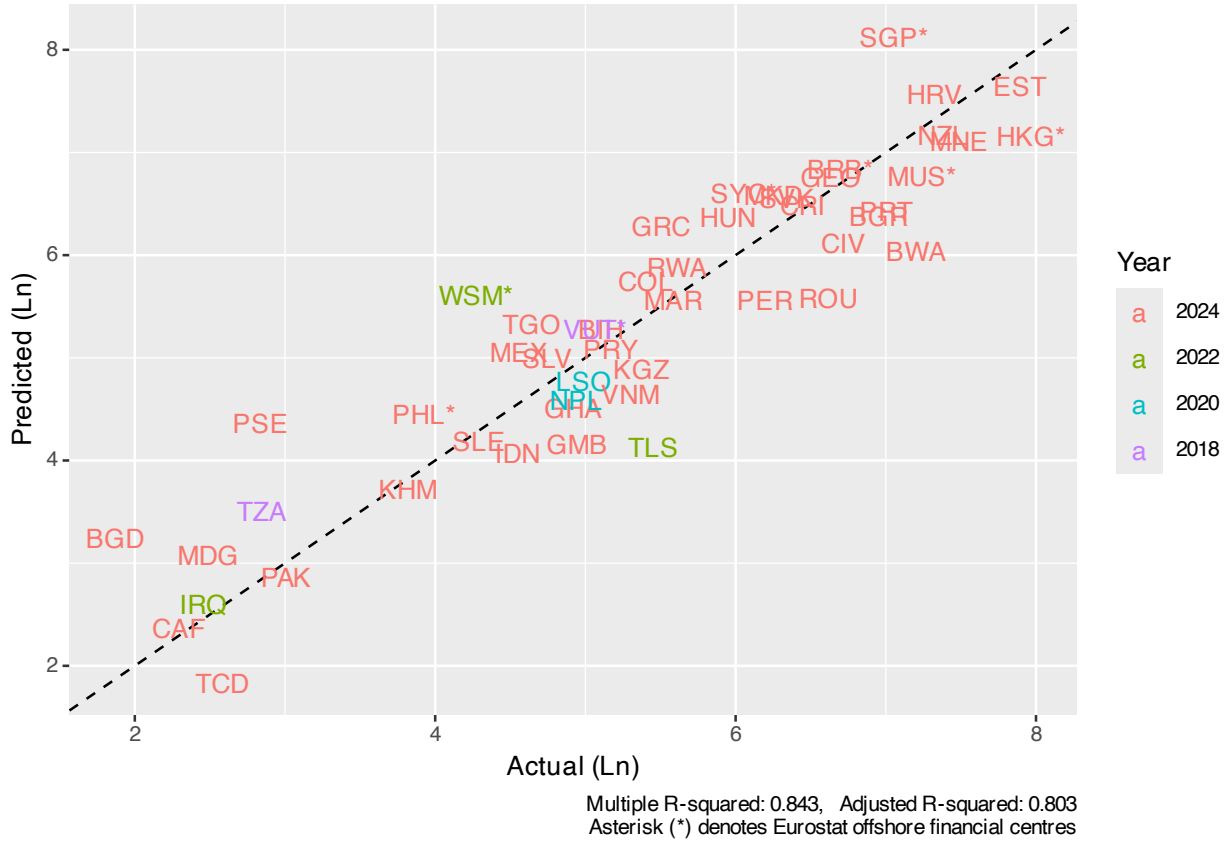
Beyond point prediction accuracy, we assess the model's ability to correctly rank countries by their entrepreneurial intensity. Spearman's rank correlation between predicted and actual values is 0.914, indicating that the model captures the ordinal relationships nearly perfectly. This high rank correlation makes the model particularly valuable for policy applications focused on relative positioning rather than absolute levels. However, Kendall's tau ($\tau = 0.746$) is somewhat lower, suggesting that extreme values – countries with very high or very low new business density – may be driving some of the correlation. This warrants caution in interpreting predictions for countries at the extremes of the distribution.

**Outliers: Hong Kong and Singapore**

Two observations deserve special mention: Hong Kong and Singapore. These city-states are global financial centers with unique characteristics that set them apart from the rest of the sample. Both have remote business registration capabilities, extraordinarily high GDP per capita, extreme population concentration, and institutional environments that differ qualitatively from other high-income countries. In terms of GDP per capita and population density, they are not comparable to any other point in the dataset.

Ideally, these observations would be studied separately given their unique characteristics. However, given the limited sample size, we cannot sacrifice them without further reducing degrees of freedom. Their inclusion is a limitation, though diagnostic plots with these observations hidden (see Figures 12 and 13 in Appendix) confirm that the model's performance does not hinge entirely on these two influential points.

Figure 11. Predicted vs. Actual (Log)




## 5.3 Variable Contribution Decomposition

Waterfall charts (see Figures 14-17 in Appendix) decompose each country's predicted new business density into the additive contributions of each independent variable, starting from the model intercept. This decomposition reveals several patterns: for countries with high entrepreneurial density (e.g., Hong Kong, Singapore, Estonia, New Zealand), the weighted average Business-Ready score and GDP per adult make substantial positive contributions, while the competition effect (adult population) and income group dummies may pull in the opposite direction. For low-density countries (e.g., Bangladesh, Central African Republic, Iraq), the institutional quality score and GDP per adult exhibit more modest contribution.

The relative contribution breakdown shows that the weighted average score is among the most influential factors for nearly all countries, confirming its central role. It is also noteworthy that the countries having new business density rate above 50th percentile exhibit a significant structural increase in weighted average score and its relative contribution (see Figure 17 in Appendix). Urbanization and working-age share provide meaningful positive contributions across the board, while capital requirements and cost effects are more variable.

An additional detailed example of variable contribution decomposition can be found in EU South European Countries Variable Contribution Decomposition section.

## 5.4 Country-to-Country Distance Decomposition

Using Mahalanobis distance – a multivariate measure that accounts for correlations between variables – we identify which countries are most similar in terms of their predictor profiles by calculating distances between vectors of variable contributions (see Table 6 in Appendix). The closest pairs include Greece and Bulgaria (distance 0.97, EU, high income), North Macedonia and Montenegro (0.996, the Balkans, non-EU, upper-middle income), El Salvador and Paraguay (1.160, Latin America, upper-middle income). These groupings largely reflect geographic and developmental similarities.

A heatmap of pairwise Mahalanobis distances (see Figure 18 in Appendix) provides a visual overview of multivariate similarity across the sample. Notable clusters include the Balkan countries (Greece, Bulgaria, Croatia, North Macedonia), the Southern African economies (Botswana, Lesotho), and the island states (Samoa, Vanuatu). Hong Kong and Singapore appear as outliers even in this multivariate space, confirming their unique position and reinforcing the need for caution in interpreting their results.

# 6 Conclusion

This paper investigates the determinants of new business formation using a novel dataset combining the World Bank's new Business Ready (B-READY) 2024 indicators with traditional macroeconomic and demographic variables. Analyzing a diverse sample of 50 countries, we find strong support for our central hypothesis: institutional quality, as measured by the Business-Ready scores, is a robust and substantively important predictor of entrepreneurial activity.

Our preferred model explains over 80% of the cross-country variation in new business density and correctly ranks countries with near-perfect accuracy (Spearman's $\rho = 0.914$). The key findings can be summarized as follows.

First, the quality of the business environment matters enormously. A one percentage point increase in our weighted average Business-Ready score is associated with a 6.5% increase in new business density, an effect that remains highly significant across all specifications. This confirms that policies improving regulatory frameworks, public services, and operational efficiency can substantially stimulate entrepreneurship.

Second, market size attracts entrepreneurs, but competition among them tempers this effect. While higher GDP per adult strongly encourages new business formation, a larger adult population – holding total economic output constant – intensifies competition and reduces entry. This nuanced finding reconciles seemingly contradictory predictions about population effects.

Third, demographic structure matters. Urbanization and working-age population share both positively influence new business density, supporting theories emphasizing "economies of density" and human capital availability. Each percentage point increase in urbanization is associated with approximately a 1.3% increase in entrepreneurial activity.

Fourth, administrative barriers discourage formal entrepreneurship. Higher costs of business registration and minimum capital requirements both show negative associations with new business density, though significance levels vary across specifications.

Fifth, income group effects persist even after controlling for these factors, suggesting that the broader development context shapes entrepreneurship in ways not fully captured by measured variables. Interestingly, higher income levels are associated with lower entrepreneurial density conditional on other factors, perhaps reflecting saturation or the transition from necessity to opportunity entrepreneurship.

Several limitations deserve acknowledgment. The sample excludes major economies lacking 2024 Business-Ready data, limiting generalizability. The inclusion of seven pre-2020 observations introduces measurement error, though diagnostic tests suggest this does not critically affect core coefficients. Multicollinearity among some predictors, particularly GDP per adult and income group dummies, inflates standard errors and warrants cautious interpretation. Finally, the unique cases of Hong Kong and Singapore suggest that city-states may follow different dynamics. Endogeneity may also be a cause of concern: GDP per capita for 2023 was tested to exclude simultaneity with the newly registered businesses in 2024, however a more rigorous approach would be greatly beneficial to further research. Some interactions were tested and hinted at varying non-linear dynamics among different income groups: countries in higher income groups attract fewer new businesses per every percentage increase in GDP per adult, but simultaneously suffer less from population increasing competition and putting strain on public services.

Despite these limitations, the analysis provides robust evidence that institutional quality, market conditions, demographic structure, and regulatory barriers jointly determine entrepreneurial outcomes. The high rank correlation makes our model particularly valuable for policy applications focused on improving countries' relative attractiveness for business formation. Future research with larger samples and panel data could further explore causal mechanisms and track how these relationships evolve over time.

# 7 Appendix

**Figure 2. Countries Distribution by GDP per Adult**

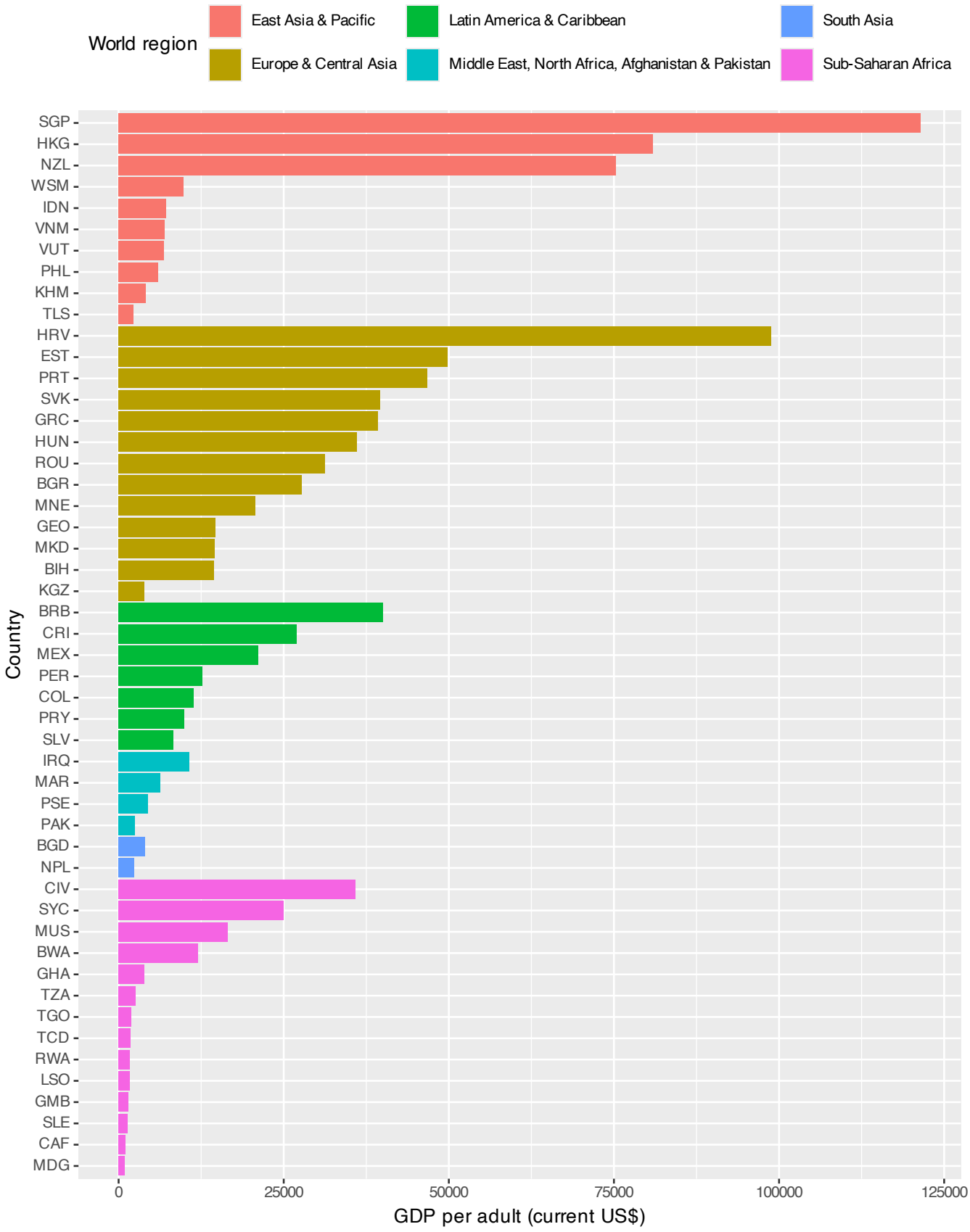


**Figure 3. Countries Distribution by New Business Density Rate**

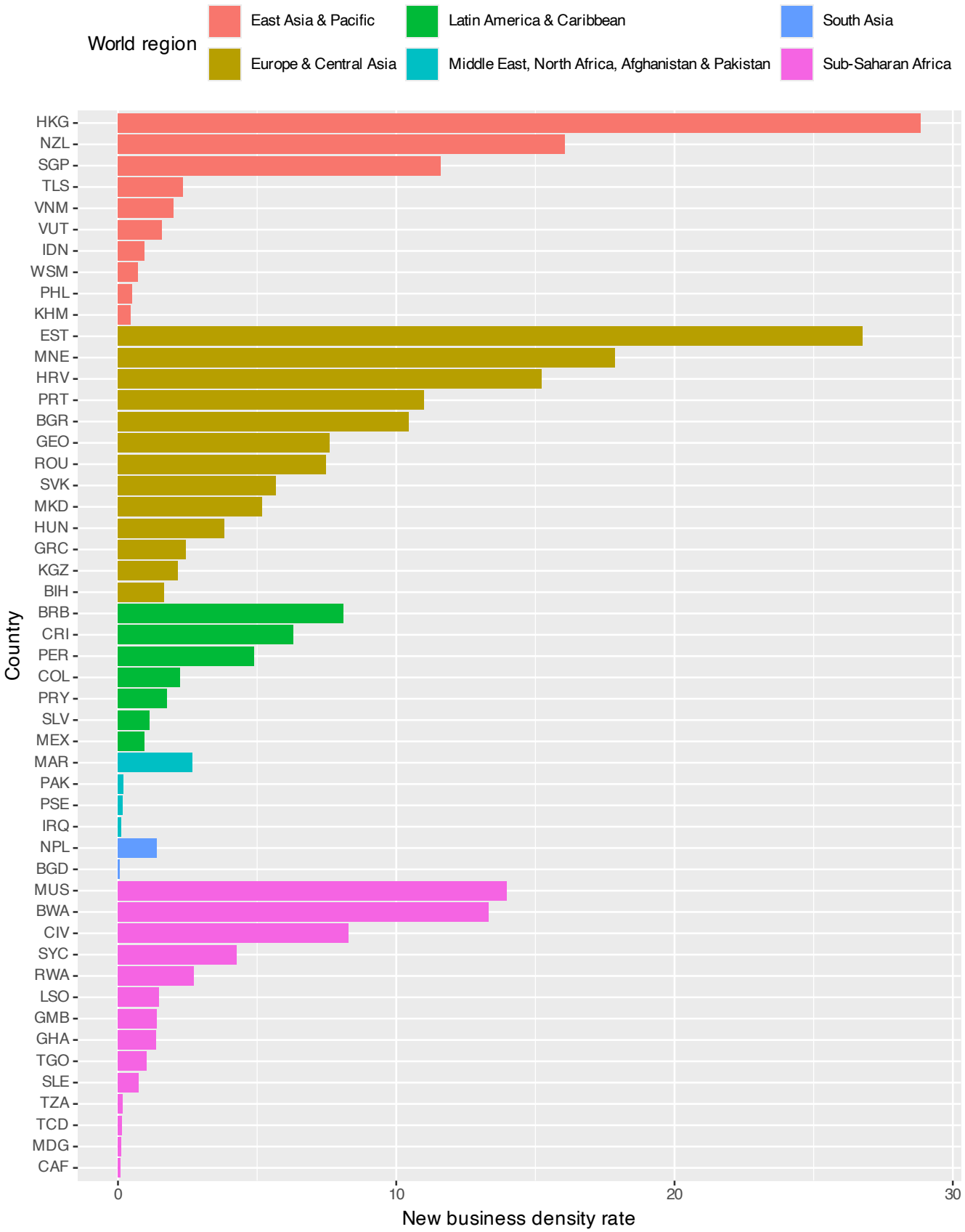
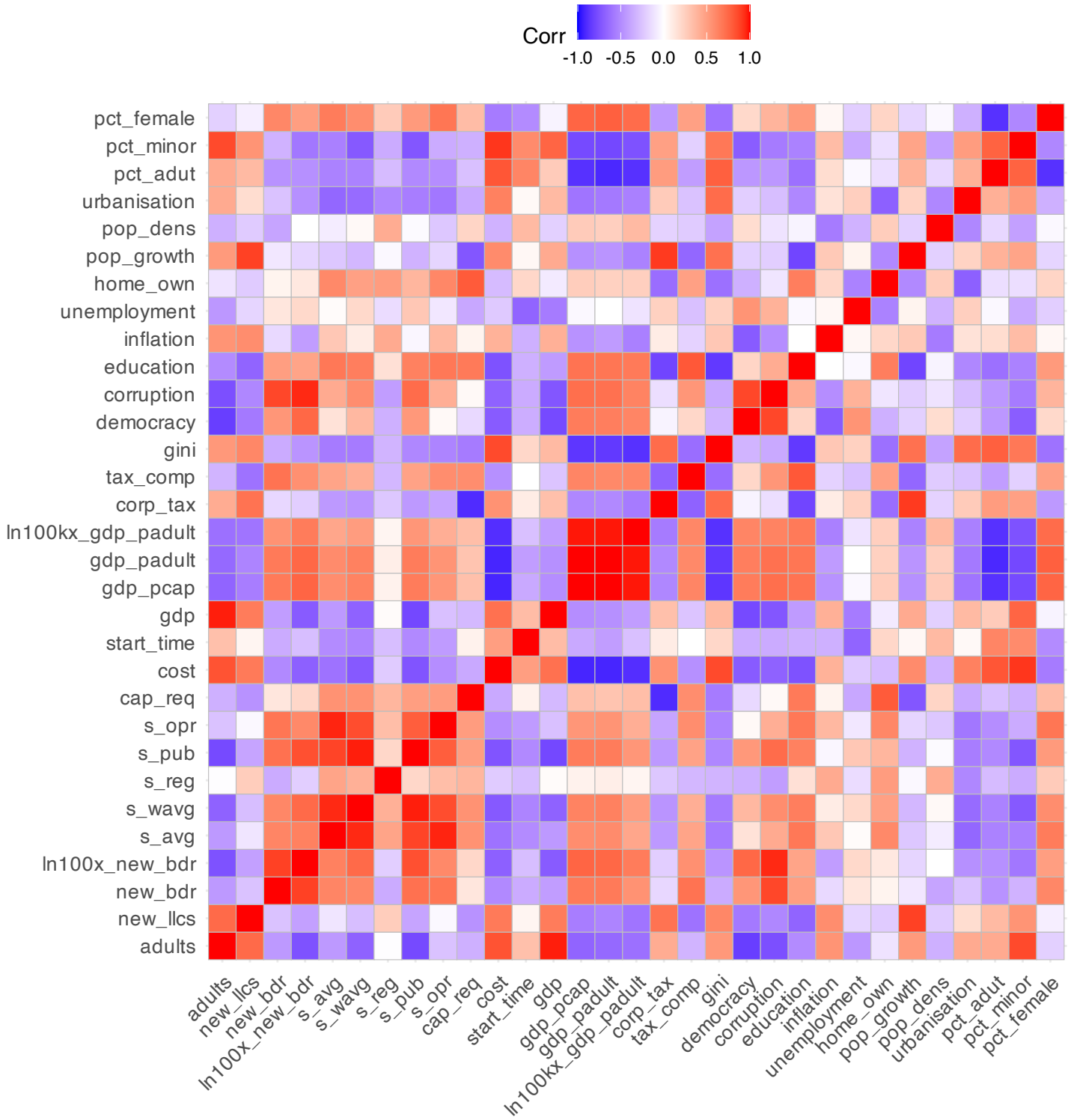


**Figure 4. Correlation Matrix**



## Table 3. Individual Effects Models

| | Weighted Avg. | Capital Requirements | Cost of Starting | Time to Start | Tax | Gini | Democracy | Corruption | Education | Inflation | Unemployment | Home Owner-ship | Urbanisation | Population Growth | Population Density | Adult % | Minor % | Female % | Offshore | Remote | Income Group | Best |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| (Intercept) | 2.048 | 2.035 | 2.964 | 1.963 | 2.425 | 2.437 | 4.100++ | 4.262++ | 1.315 | 2.383 | 2.672 | 2.654 | 4.828++ | 2.658 | 2.041 | 1.449 | 5.385+ | 0.358 | 1.847 | 2.258 | -3.018 | -2.780 |
| | (3.005) | (2.930) | (2.887) | (3.058) | (3.071) | (3.555) | (2.851) | (2.766) | (3.167) | (3.212) | (3.085) | (3.090) | (3.346) | (3.096) | (3.054) | (2.940) | (4.169) | (5.091) | (3.021) | (3.051) | (4.634) | (4.692) |
| log(gdp_padult * 100000) | 0.277* | 0.284* | 0.261+++ | 0.284* | 0.255+++ | 0.243++ | 0.254+++ | 0.156+ | 0.339* | 0.268+++ | 0.257+++ | 0.284* | 0.136 | 0.257+++ | 0.278+++ | 0.176+ | 0.165 | 0.280* | 0.308* | 0.267+++ | 0.555* | 0.467* |
| | (0.136) | (0.133) | (0.130) | (0.140) | (0.141) | (0.154) | (0.132) | (0.132) | (0.159) | (0.144) | (0.138) | (0.137) | (0.156) | (0.139) | (0.140) | (0.143) | (0.168) | (0.138) | (0.141) | (0.139) | (0.237) | (0.223) |
| log(adults) | -0.428*** | -0.420*** | -0.408*** | -0.426*** | -0.413*** | -0.437*** | -0.519*** | -0.429*** | -0.448*** | -0.430*** | -0.434*** | -0.427*** | -0.472*** | -0.417*** | -0.428*** | -0.454*** | -0.425*** | -0.428*** | -0.446*** | -0.424*** | -0.454*** | -0.489*** |
| | (0.073) | (0.071) | (0.070) | (0.074) | (0.077) | (0.075) | (0.081) | (0.079) | (0.078) | (0.077) | (0.074) | (0.073) | (0.076) | (0.074) | (0.075) | (0.072) | (0.073) | (0.074) | (0.076) | (0.074) | (0.076) | (0.077) |
| s_wavg | 7.496*** | 7.202*** | 6.221*** | 7.427*** | 7.650*** | 7.493*** | 6.570*** | 6.976*** | 7.877*** | 7.329*** | 7.340*** | 7.579*** | 7.657*** | 7.049*** | 7.494*** | 7.054*** | 6.760*** | 7.336*** | 7.305*** | 7.364*** | 8.009*** | 6.543*** |
| | (1.376) | (1.352) | (1.415) | (1.421) | (1.402) | (1.432) | (1.384) | (1.362) | (1.470) | (1.437) | (1.389) | (1.383) | (1.350) | (1.475) | (1.396) | (1.359) | (1.513) | (1.442) | (1.397) | (1.407) | (1.425) | (1.455) |
| cap_req | | -1.618+++ (0.881) | | | | | | | | | | | | | | | | | | | | -0.449 (0.954) |
| cost | | | -0.010* (0.004) | | | | | | | | | | | | | | | | | | | -0.008++ (0.005) |
| start_time | | | | -0.001 (0.004) | | | | | | | | | | | | | | | | | | |
| corp_tax | | | | | -1.089 (1.574) | | | | | | | | | | | | | | | | | |
| gini | | | | | | 1.211 (1.953) | | | | | | | | | | | | | | | | |
| democracy | | | | | | | 0.830+ (0.783) | | | | | | | | | | | | | | | |
| corruption | | | | | | | | 1.522++ (0.984) | | | | | | | | | | | | | | |
| education | | | | | | | | | -0.050 (0.066) | | | | | | | | | | | | | |
| inflation | | | | | | | | | | -0.391 (1.360) | | | | | | | | | | | | |
| unemployment | | | | | | | | | | | 0.072 (1.936) | | | | | | | | | | | |
| home_own | | | | | | | | | | | | -1.024 (1.160) | | | | | | | | | | |
| urbanisation | | | | | | | | | | | | | 1.274+++ (0.732) | | | | | | | | | 1.299+++ (0.718) |
| pop_growth | | | | | | | | | | | | | | -10.966 (12.764) | | | | | | | | |
| log(pop_dens) | | | | | | | | | | | | | | | -0.002 (0.082) | | | | | | | |
| pct_adut | | | | | | | | | | | | | | | | 5.301+++ (2.784) | | | | | | 4.597++ (3.060) |
| pct_minor | | | | | | | | | | | | | | | | | -2.410+ (2.094) | | | | | |
| pct_female | | | | | | | | | | | | | | | | | | 3.369 (8.152) | | | | |
| is_ofcTRUE | | | | | | | | | | | | | | | | | | | -0.304 (0.346) | | | |
| remoteTRUE | | | | | | | | | | | | | | | | | | | | 0.252 (0.454) | | |
| g_incomeLower middle income | | | | | | | | | | | | | | | | | | | | | -0.178 (0.443) | -0.655++ (0.450) |
| g_incomeUpper middle income | | | | | | | | | | | | | | | | | | | | | -0.630 (0.607) | -1.345* (0.603) |
| g_incomeHigh income | | | | | | | | | | | | | | | | | | | | | -1.208++ (0.863) | -1.884* (0.821) |
| Num.Obs. | 50 | 50 | 50 | 50 | 50 | 46 | 46 | 47 | 50 | 47 | 49 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 | 50 |
| R2 | 0.774 | 0.790 | 0.799 | 0.774 | 0.776 | 0.758 | 0.823 | 0.827 | 0.777 | 0.765 | 0.764 | 0.778 | 0.788 | 0.777 | 0.774 | 0.791 | 0.780 | 0.775 | 0.778 | 0.775 | 0.787 | 0.843 |
| R2 Adj. | 0.759 | 0.771 | 0.781 | 0.754 | 0.756 | 0.734 | 0.805 | 0.810 | 0.757 | 0.743 | 0.743 | 0.758 | 0.769 | 0.758 | 0.754 | 0.772 | 0.761 | 0.755 | 0.758 | 0.755 | 0.758 | 0.803 |
| AIC | 195.2 | 193.6 | 191.2 | 197.1 | 196.7 | 172.7 | 172.0 | 177.1 | 196.6 | 193.6 | 198.7 | 196.4 | 194.0 | 196.4 | 197.2 | 193.3 | 195.8 | 197.0 | 196.4 | 196.9 | 198.1 | 190.9 |
| BIC | 204.8 | 205.1 | 202.7 | 208.6 | 208.2 | 183.7 | 183.0 | 188.2 | 208.0 | 204.7 | 210.1 | 207.8 | 205.4 | 207.9 | 208.7 | 204.8 | 207.2 | 208.5 | 207.8 | 208.3 | 213.4 | 213.8 |
| Log.Lik. | -56.922 | -55.114 | -53.932 | -56.891 | -56.657 | -52.485 | -48.007 | -46.900 | -56.599 | -54.248 | -55.379 | -56.493 | -55.292 | -56.515 | -56.921 | -54.984 | -56.197 | -56.827 | -56.497 | -56.751 | -55.359 | -47.764 |
| F | 52.436 | 42.201 | 44.788 | 38.534 | 39.001 | | | | 39.118 | | | 39.333 | 41.821 | 39.288 | 38.473 | 42.478 | 39.935 | 38.661 | 39.324 | 38.812 | 26.551 | 20.963 |
| RMSE | 0.76 | 0.73 | 0.71 | 0.75 | 0.75 | 0.76 | 0.69 | 0.66 | 0.75 | 0.77 | 0.75 | 0.75 | 0.73 | 0.75 | 0.76 | 0.73 | 0.74 | 0.75 | 0.75 | 0.75 | 0.73 | 0.63 |

+ p < 0.3, ++ p < 0.2, +++ p < 0.1, * p < 0.05, ** p < 0.01, *** p < 0.001

**Table 5. Best Model Multicollinearity Assessment: Linear Restriction Tests**

| term | estimated_coeff | hypothetical_coeff | Pr(>F) |
|---|---|---|---|
| (Intercept) | -2.780 | 2.336 | 0.282 |
| log(gdp_padult * 100000) | 0.467 | 0.266 | 0.375 |
| log(adults) | -0.489 | -0.437 | 0.503 |
| s_wavg | 6.543 | 7.295 | 0.608 |
| cap_req | -0.449 | -1.618 | 0.228 |
| cost | -0.008 | -0.010 | 0.694 |
| urbanisation | 1.299 | 1.274 | 0.973 |
| pct_adut | 4.597 | 5.301 | 0.819 |
| g_incomeLower middle income | -0.655 | -0.178 | 0.295 |
| g_incomeUpper middle income | -1.345 | -0.630 | 0.243 |
| g_incomeHigh income | -1.884 | -1.208 | 0.415 |

**Figure 7. Semi-Partial R²**

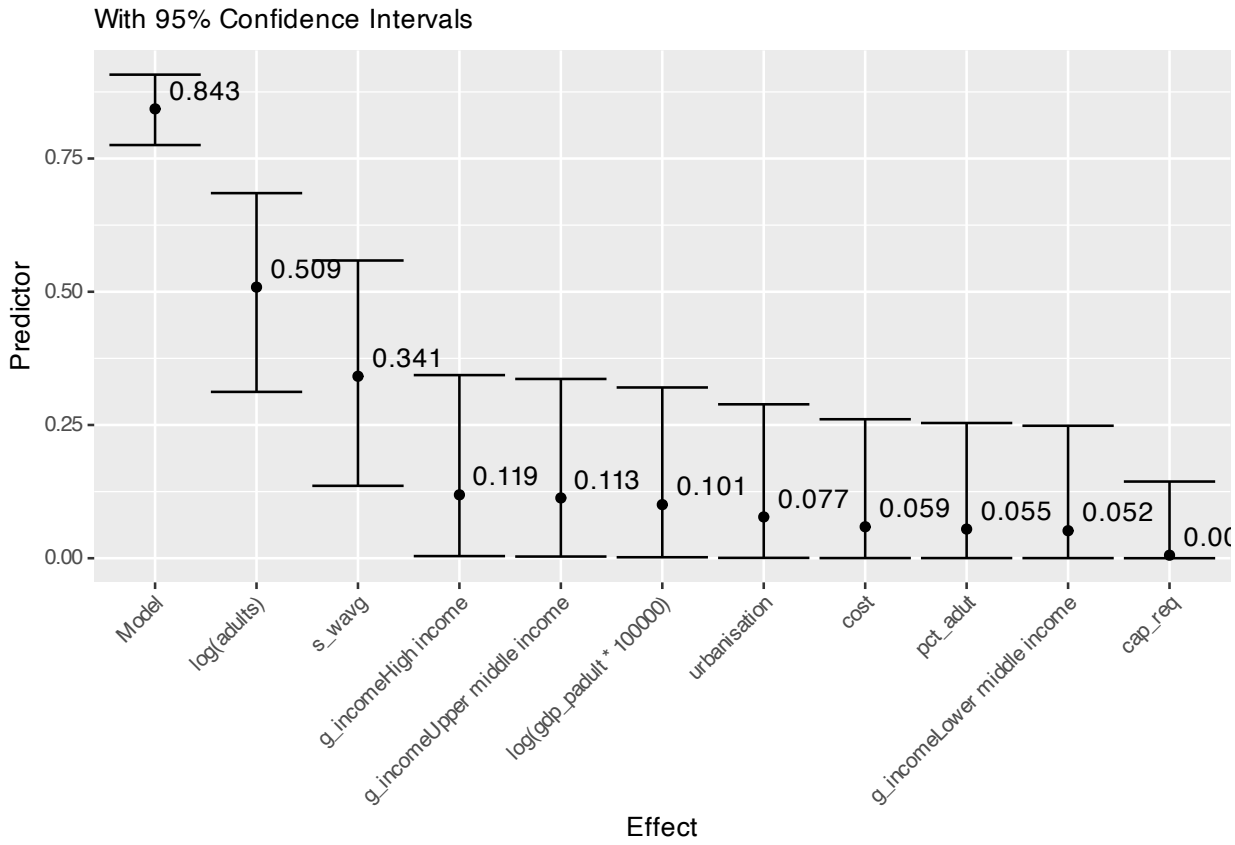



**Figure 8. Standardized Coefficients**

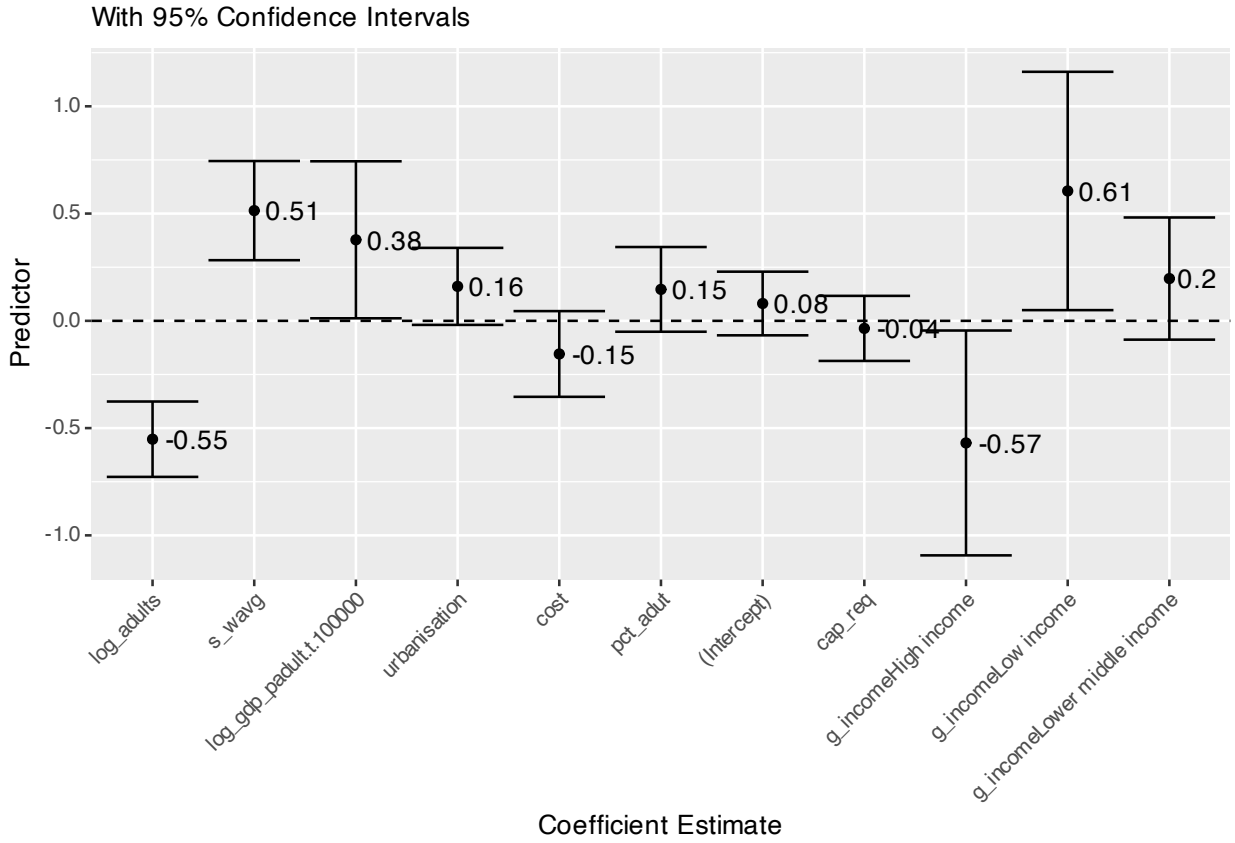


**Figure 9. Residuals Q-Q Plot**

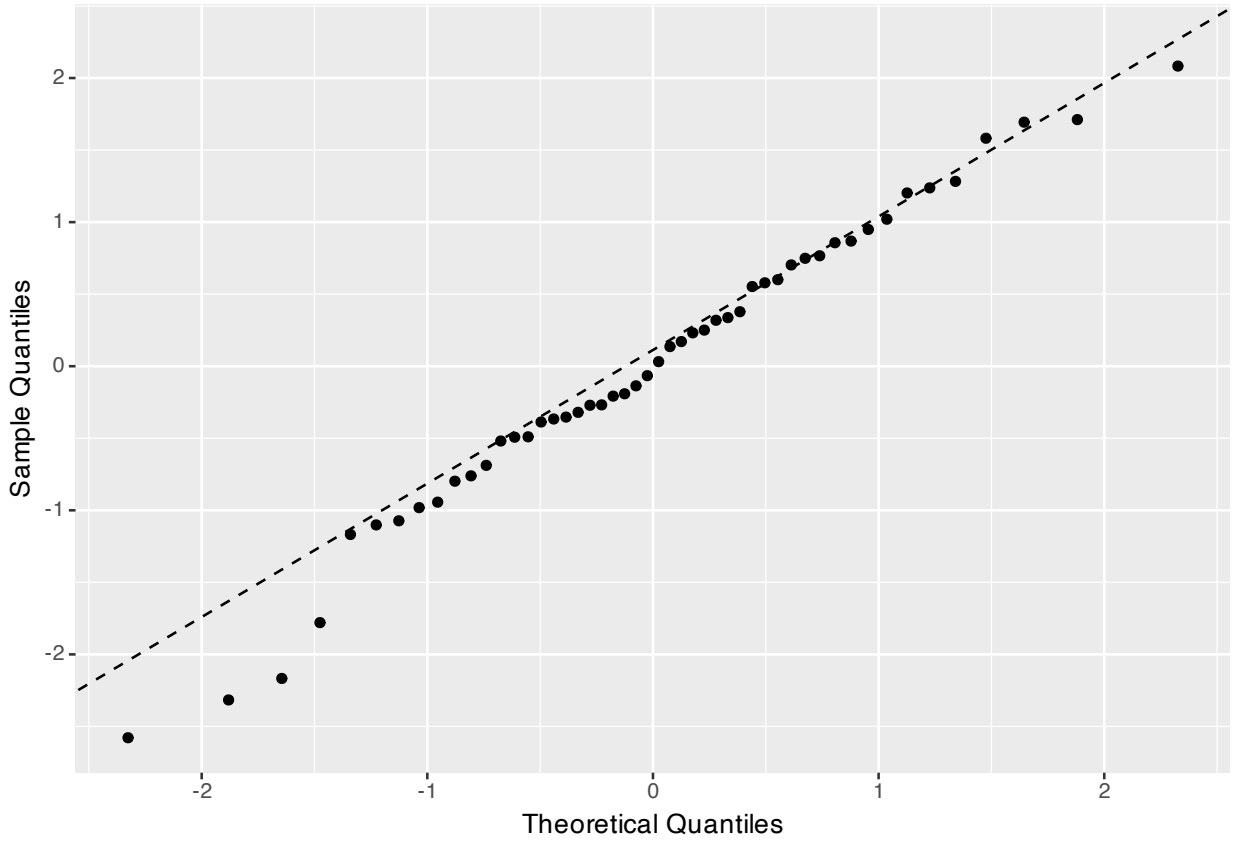



**Figure 10. Fitted Valus against Residuals**

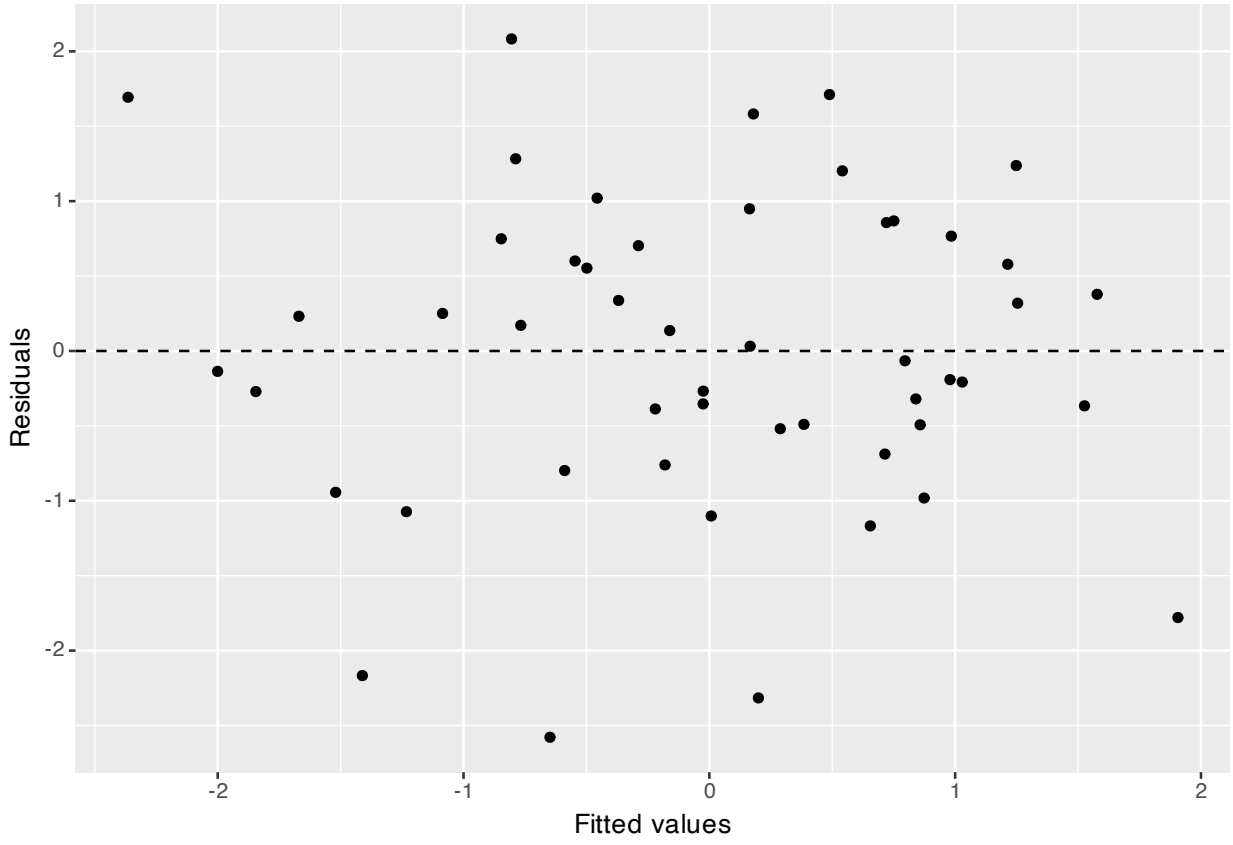
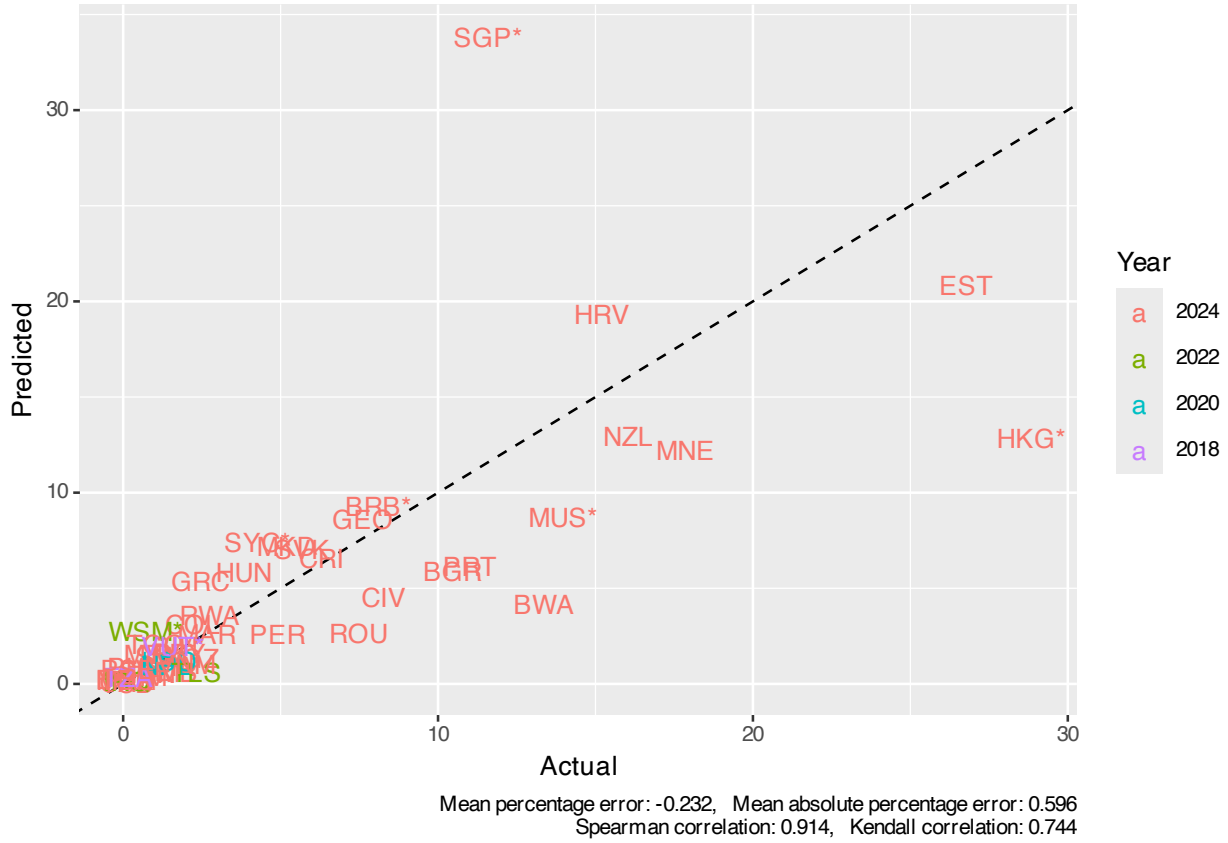


**Figure 12. Predicted vs. Actual**




Mean percentage error: -0.232, Mean absolute percentage error: 0.596
Spearman correlation: 0.914, Kendall correlation: 0.744

**Figure 13. Predicted vs. Actual (Hong Kong and Singapore Hidden)**




Mean percentage error: -0.232, Mean absolute percentage error: 0.596
Spearman correlation: 0.914, Kendall correlation: 0.744

**Figure 14. Contribution Breakdown per Prediction**

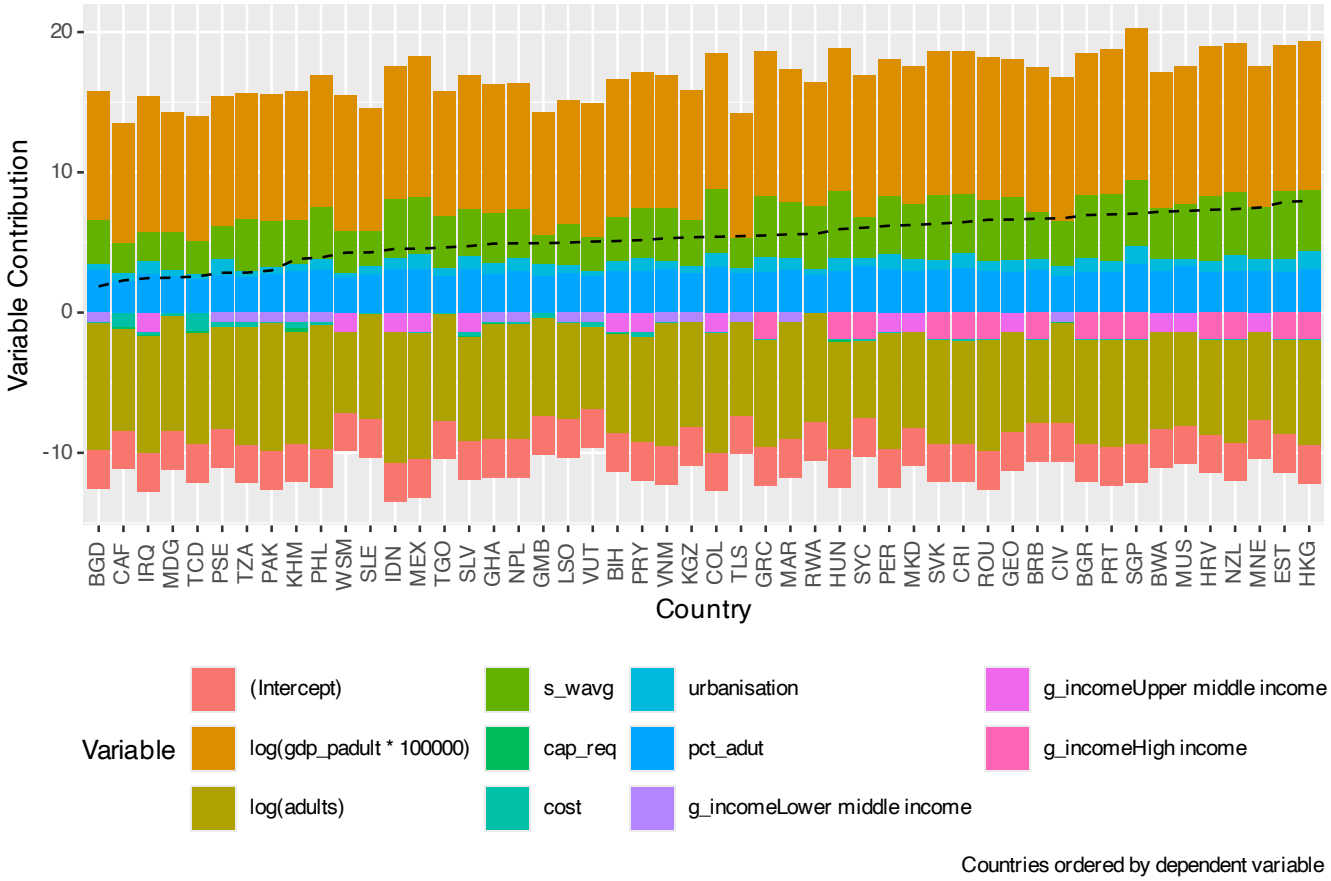



**Figure 15. Relative Contribution Breakdown per Prediction**

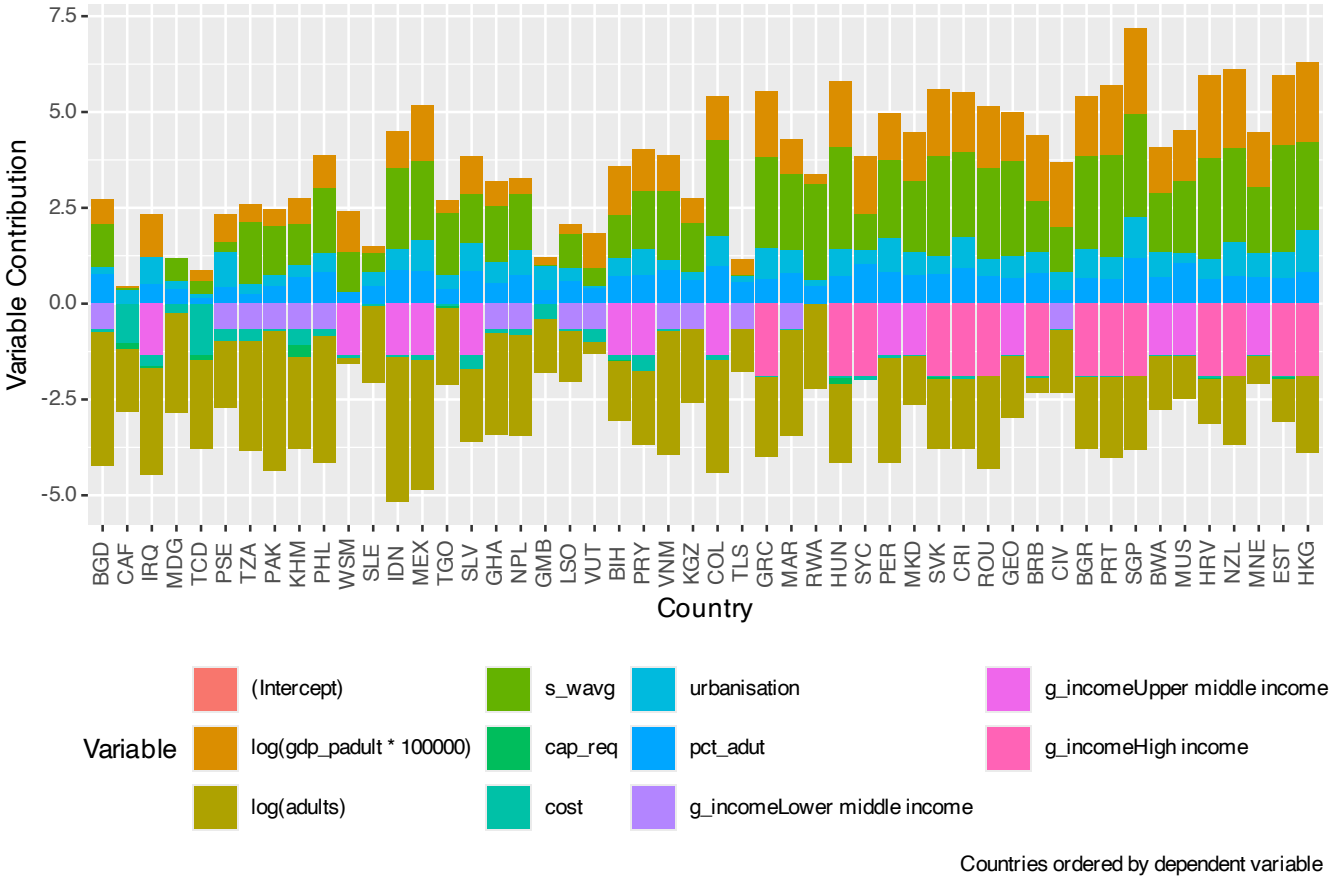


**Figure 16. Contribution Breakdown Trend**

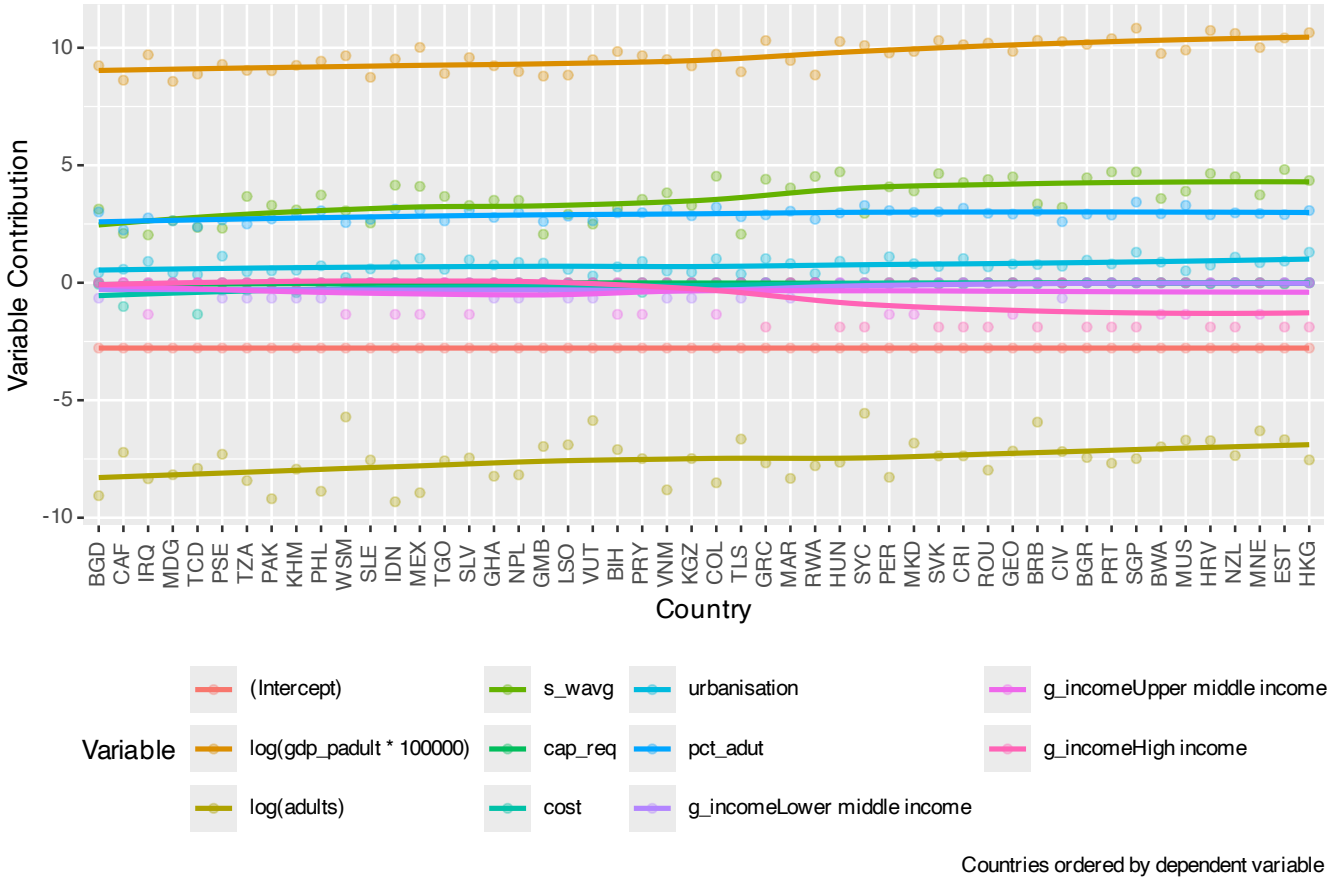



**Figure 17. Relative Contribution Breakdown Trend**

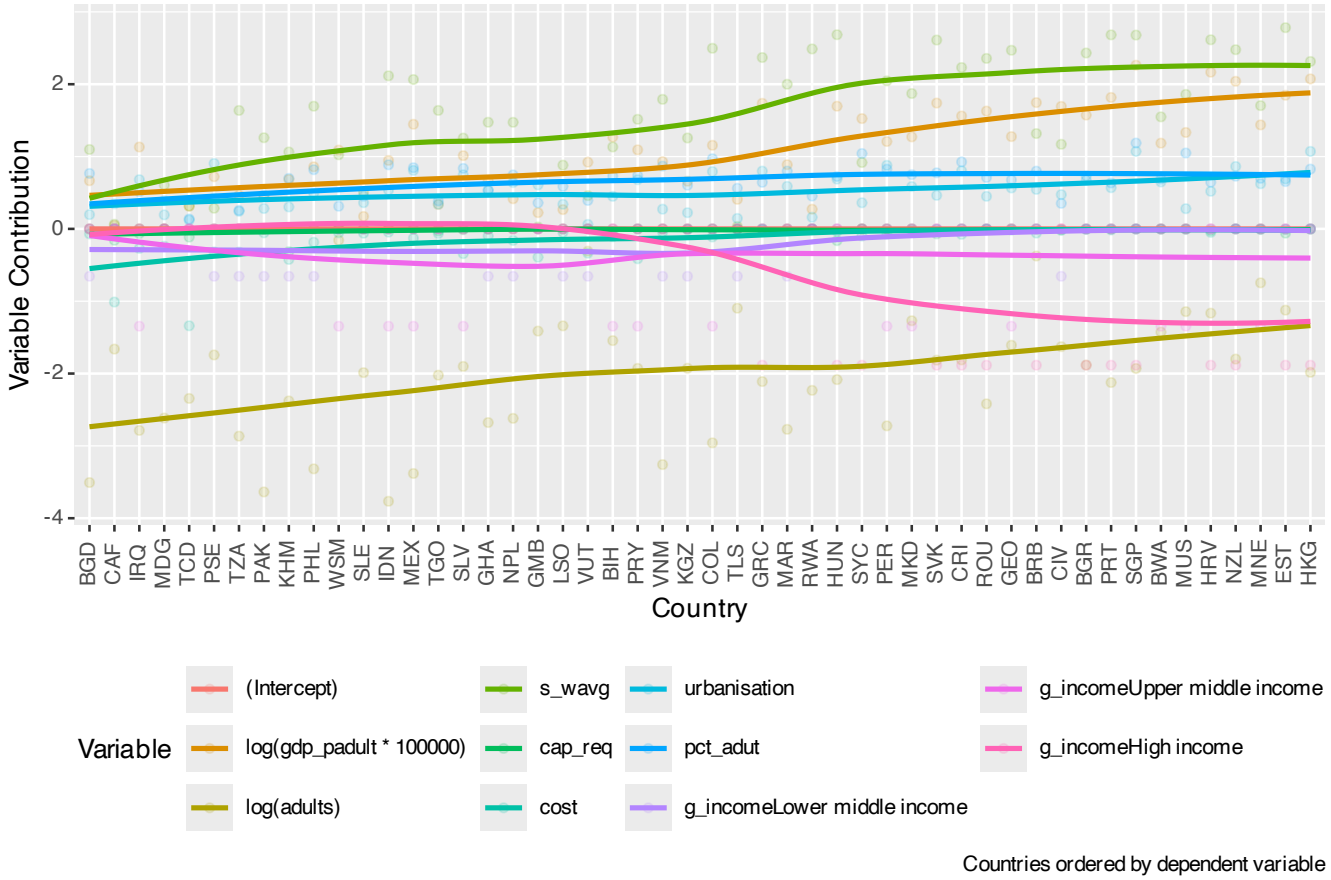
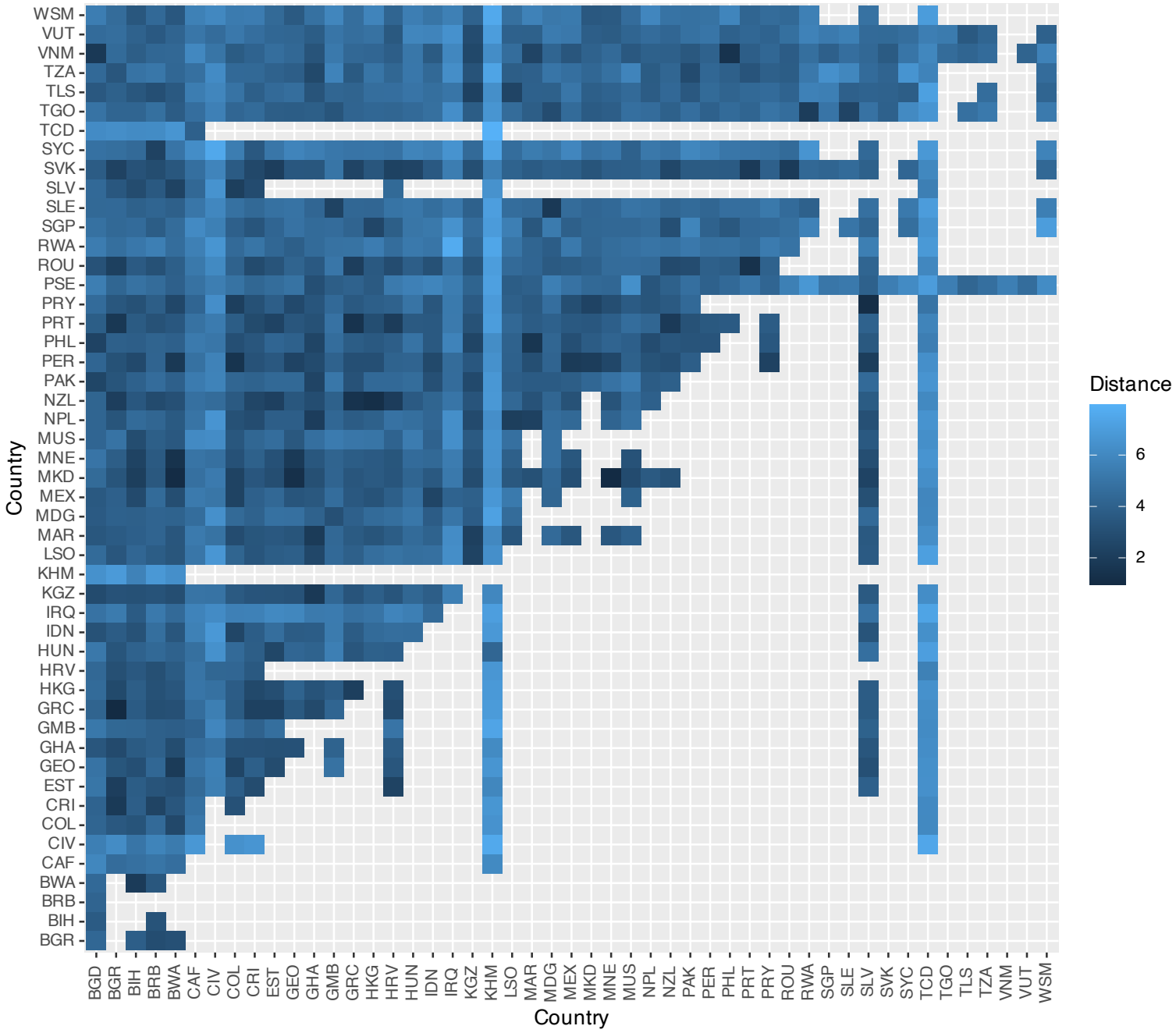


**Figure 18. Country-to-Country Mahalanobis Distance Heatmap**



## 7.1 EU South European Countries Variable Contribution Decomposition

Continuing the logic outlined in Variable Contribution Decomposition and Country-to-Country Distance Decomposition sections, we can proceed to compare individual countries in greater detail. The 10 most similar pairs of countries identified using Mahalanobis distance are listed in in Table 6. It is noteworthy, that a cluster of Southern European countries (BGR, GRC, MNE, GEO, MKD, ROU, PRT) is immediately identified.

**Table 6. 10 Closest Country Pairs**

| item1 | item2 | distance |
|---|---|---|
| BGR | GRC | 0.970 |
| MNE | MKD | 0.996 |
| BWA | MKD | 1.037 |
| SLV | PRY | 1.160 |
| HKG | NZL | 1.192 |
| GEO | MKD | 1.281 |
| PRT | ROU | 1.379 |
| BWA | MNE | 1.428 |
| GRC | PRT | 1.450 |
| GRC | NZL | 1.452 |

For demonstration purposes we proceed to compare BGR, GRC, HUN, HRV and PRT, each being a high income country and an EU member (see Table 7). Actual and predicted new business density rate is remarkably well aligned, preserving the rank and making it possible to make meaningful comparison on variable contribution level.

**Table 7. South EU Countries**

| code | country | w_region | block | g_income | g_lending | ln100x_new_bdr | fitted.values |
|---|---|---|---|---|---|---|---|
| HRV | Croatia | Europe & Central Asia | EU | High income | IBRD | 7.33 | 7.57 |
| PRT | Portugal | Europe & Central Asia | EU | High income | NA | 7.00 | 6.42 |
| BGR | Bulgaria | Europe & Central Asia | EU | High income | IBRD | 6.95 | 6.38 |
| HUN | Hungary | Europe & Central Asia | EU | High income | NA | 5.95 | 6.37 |
| GRC | Greece | Europe & Central Asia | EU | High income | NA | 5.50 | 6.28 |

Relative contribution plot (Figure 20) is particularly revealing: while most of the difference appear to be coming from the combination of weighted average score and GDP per adult, some particular aspects are making it nuanced, HRV experiencing the least (and almost twice smaller than the other four countries) pressure from the population and HUN exhibiting exceptionally high weighted average score despite lacking in GDP per adult and substantial capital requirements. Up until PRT (the second highest among the five by new business density) higher GDP per adult does not coincide with higher business density rate, while higher urbanisation rate does not coincide with higher business density rate in four cases out of five, both insights stressing the importance of interaction of factors.

**Figure 19. Contribution Breakdown per Prediction (South EU)**

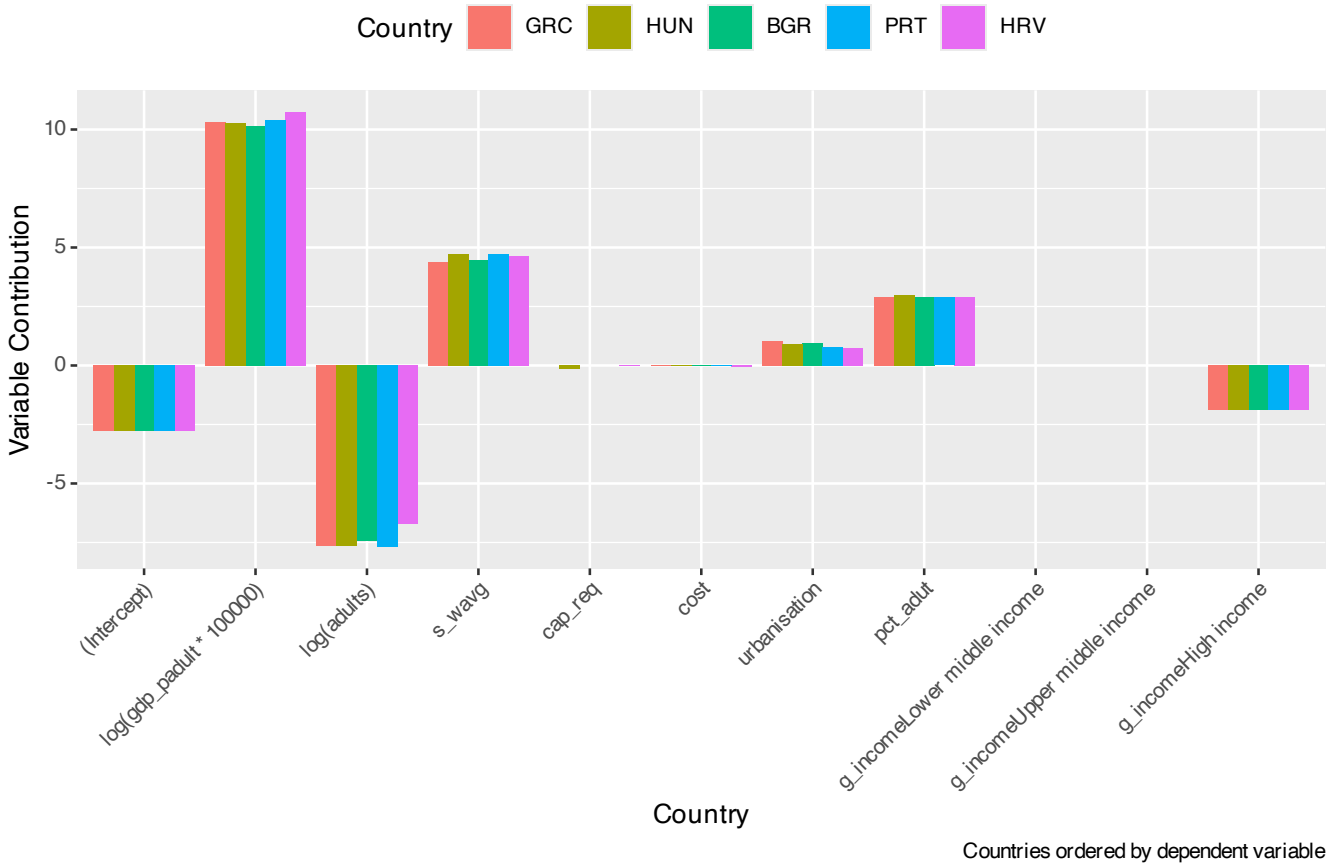
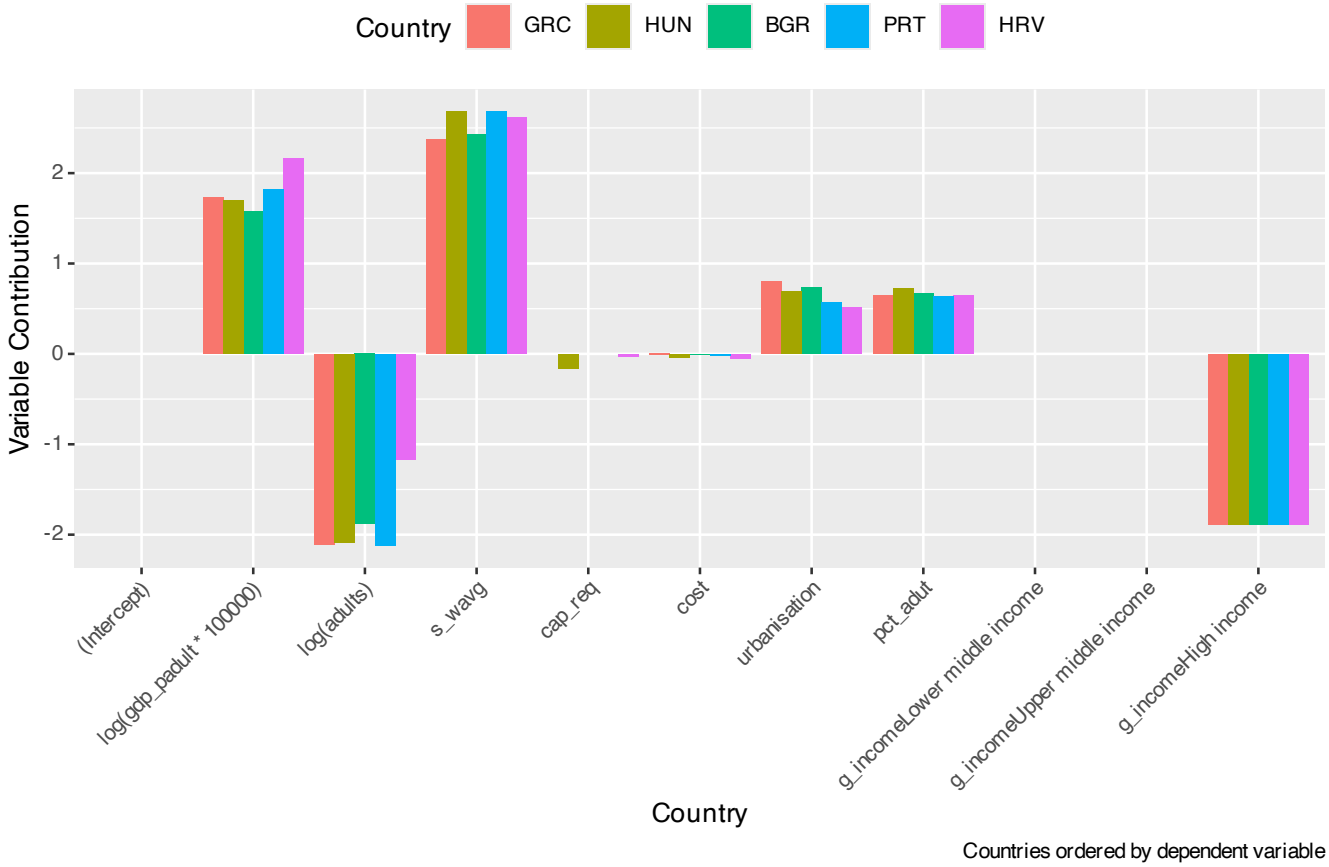



**Figure 20. Relative Contribution Breakdown per Prediction (South EU)**



## 7.2 Other Variable Contribution Decomposition Applications

The method outlined can be used for general exploratory data analysis in a new context appearing after the model specification. Some of the examples can be found in Tables 8-10, such as Latin America & Caribbean region's possible leverage of the abundant working-age force and considerably high average weighted score.

Partial dependence style plots overlaid on actual observations (see Figure Figures 16 and 17) reveal additional detail, which is not necessarily evident from the pre-modelling analysis due to lack of focus provided by the model.

Other common analytical methods such as clusterisation (partly covered in Country-to-Country Distance Decomposition section), structural break/transition detection (e.g. calculating a level increase in a cpecific variable in a certain group of observations) and advanced visualisation techniques are topics for further research.

Table 8. Average Relative Variable Contribution by Percentile of New Business Density Rate

| variable | [0-25%] | (25-50%] | (50-75%] | (75-100%] |
|---|---|---|---|---|
| (Intercept) | 0.000 | 0.000 | 0.000 | 0.000 |
| cap_req | -0.050 | -0.008 | -0.015 | -0.012 |
| cost | -0.350 | -0.162 | -0.031 | -0.028 |
| g_income | -0.828 | -0.906 | -1.465 | -1.718 |
| log(adults) | -2.446 | -2.186 | -2.043 | -1.290 |
| log(gdp_padult * 100000) | 0.479 | 0.823 | 1.328 | 1.714 |
| pct_adut | 0.455 | 0.674 | 0.649 | 0.827 |
| s_wavg | 0.693 | 1.386 | 2.108 | 2.142 |
| urbanisation | 0.379 | 0.466 | 0.565 | 0.650 |

Table 9. Average Relative Variable Contribution by Income Group

| variable | Low income | Lower middle income | Upper middle income | High income |
|---|---|---|---|---|
| (Intercept) | 0.000 | 0.000 | 0.000 | 0.000 |
| cap_req | -0.043 | -0.022 | -0.009 | -0.023 |
| cost | -0.444 | -0.142 | -0.115 | -0.027 |
| log(adults) | -2.039 | -2.338 | -1.953 | -1.616 |
| log(gdp_padult * 100000) | 0.195 | 0.713 | 1.204 | 1.815 |
| pct_adut | 0.311 | 0.592 | 0.751 | 0.783 |
| s_wavg | 0.807 | 1.172 | 1.650 | 2.319 |
| urbanisation | 0.305 | 0.380 | 0.591 | 0.688 |
| g_income | NA | -0.655 | -1.345 | -1.884 |

Table 10. Average Relative Variable Contribution by World Region

| variable | East Asia & Pacific | Europe & Central Asia | Latin America & Caribbean | Middle East, North Africa, Afghanistan & Pakistan | South Asia | Sub-Saharan Africa |
|---|---|---|---|---|---|---|
| (Intercept) | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| cap_req | -0.032 | -0.028 | -0.002 | -0.016 | 0.000 | -0.023 |
| cost | -0.110 | -0.024 | -0.172 | -0.168 | -0.114 | -0.265 |
| g_income | -1.162 | -1.624 | -1.499 | -0.828 | -0.655 | -1.028 |
| log(adults) | -1.999 | -1.678 | -2.154 | -2.734 | -3.063 | -1.811 |
| log(gdp_padult * 100000) | 1.221 | 1.548 | 1.317 | 0.798 | 0.541 | 0.607 |
| pct_adut | 0.728 | 0.686 | 0.848 | 0.552 | 0.756 | 0.478 |
| s_wavg | 1.565 | 2.228 | 1.847 | 0.886 | 1.286 | 1.081 |
| urbanisation | 0.482 | 0.566 | 0.751 | 0.614 | 0.418 | 0.358 |